Program overview

20-May-2025 12:58

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master SC 2024	**Master Systems and Control 2024**		
S&C Obligatory Courses (35 ECTS)			
SC42015	Control Theory	6	
SC42025	Filtering & Identification	6	
SC42035	Integration Project Systems and Control	5	
SC42056	Optimization for Systems and Control	3	
SC42095	Digital Control	3	
SC42145	Robust Control	3	
SC42150	Statistical Signal Processing	3	
SC42155	Modelling of Dynamical Systems	3	
SC42165	Machine Learning for Systems and Control	3	
S&C Obligatory Social Courses: choose 1 out of 3 (3 ECTS)			
WM0320TU	Ethics and Engineering	3	
WM0349WB	Philosophy of Engineering Science and Design	3	
WM0801TU	Introduction to Safety Science	3	
S&C Elective Courses (min. 22 ECTS)			
AE4301	Automatic Flight Control System Design	3	
AE4311	Nonlinear and Adaptive Flight Control	4	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4314-21	Helicopter Performance, Stability and Control	4	
AE4317	Autonomous Flight of Micro Air Vehicles	4	
AE4323	Real-time Distributed Flight and Space Simulation	3	
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4	
AE4W09	Wind Turbine Design	5	
AP3122	Advanced Optical Imaging	6	
AP3132	Advanced Digital Image Processing	6	
AP3382	Advanced Photonics	6	
AP3391	Geometrical Optics	6	
AP3401	Introduction to Charged Particle Optics	6	
CESE4025	Real-time Systems	5	
CIEQ6002	Transport Modelling and Analysis	5	
CIEQ6003	Traffic Modelling and Management	6	
CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4	
CIEQ6224	Urban and motorway Traffic Flow Modelling and Control	5	
CIEQ6231	Public Transport System and Supply Planning and Operations	5	
CIEQ6232	Public Transport Demand and Network Planning and Operations	5	
CIEQ6233	Railway Operations and Control	5	
ET4257	Sensors and Actuators	4	
ET4291	Control of Electrical Drives	5	
ME41006	Musculoskeletal Modeling and Simulation	4	
ME41056	Multibody Dynamics	5	
ME45155	Computational Fluid Dynamics for Engineers	5	
ME46010	Intro to Nanoscience and Technology	3	
ME46055	Engineering Dynamics	4	
ME46085-23	Mechatronic System Design	4	
RO47003	Robot Software Practicals	5	
RO47004	Machine Perception	5	
RO47005	Planning & Decision Making	5	
RO47017	Vehicle Dynamics and Control	5	
RO47019	Intelligent Control Systems	4	
SC42030	Control for High Resolution Imaging	3	
SC42061	Nonlinear Control Systems	3	
SC42065	Adaptive Optics Design Project	3	
SC42075	Modeling and Control of Hybrid Systems	3	
SC42101	Networked and Distributed Control Systems	4	
SC42110	Dynamic Programming and Stochastic Control	5	
SC42115	Internship	6	
SC42125	Model Predictive Control	4	
SC42130	Fault Diagnosis and Fault Tolerant Control	4	
SC42140	Signal Analysis & Learning for Two-Dimensional Systems	3	
SC42160	Data Compression: Entropy and Sparsity Perspectives	5	
SC42170	Probabilistic Sensor Fusion	3	

SC47500	Tensor Networks for Green AI and Signal Processing	4
WI4062TU	Transport, Routing and Scheduling	3
WI4201	Scientific Computing	6
WI4212	Advanced Numerical Methods	6
WI4221	Control of Discrete-Time Stochastic Systems	6
WI4260TU	Scientific Programming for Engineers	3
WI4620	Semidefinite Optimization	6

S&C YEAR 2

S&C Graduation Project Obligatory (45 ECTS)

SC52010	S&C Literature Research	10
SC52135	S&C MSc Thesis	35

S&C Elective Course Select 1 out of 3: joint interdiscipinary project, or research project + min 5 EC additional electives, or min 15 EC additional electives

IFM4040	Joint Interdisciplinary Project	15
SC52055	Research Assignment	10

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

Master SC 2024	
Director of Studies	Dr.ir. A.J.J. van den Boom
Program Title	Master Systems & Control
Program Coordinator	The MSc coordinator can be contacted for questions or problems related to the individual study programme and for monitoring progress. E-mail: sc-coordinator-dcsc@tudelft.nl.
In association with the University of	Relationship with 4TU graduate school The MSc programme in Systems and Control at Delft University of Technology is part of the 4TU MSc programme in Systems and Control. The other participating groups in this programme are: Control Systems Technology Group, Department of Mechanical Engineering, Eindhoven University of Technology, Control Systems Group, Department of Electrical Engineering, Eindhoven University of Technology, The Systems & Control porgramme at the university of Twente has been discontinued. The University of Wageningen is not involved in the 4TU MSc programme in Systems and Control. Relationship with the national graduate school, DISC The MSc is an excellent way of preparing for the PhD programme offered by the national graduate school, DISC (Dutch Institute of Systems and Control). This is housed in the same research center at Delft University of Technology. http://www.disc.tudelft.nl
Introduction 1	The MSc programme in Systems and Control began in September 2003. It is taught by the Delft Center for Systems and Control (DCSC) within the Faculty of Mechanical Engineering. For more information, visit www.dcsc.tudelft.nl.
Program Goals	The MSc programme in Systems and Control covers the analysis and design of reliable and high-performance measurement and control strategies for a wide variety of dynamic technological processes. It centres on fundamental generic aspects of systems and control engineering and stresses the multidisciplinary nature of the field, with applications in mechanical engineering, electrical engineering, applied physics, aerospace engineering, and chemical engineering among them the following: - High-accuracy positioning and motion-control systems, mechatronics, micro-systems, production systems, robotics, and smart structures; - Petrochemical, chemical, physical, and biotechnological production processes; - Transportation systems (automotive, railway, logistics, aerospace); - Infrastructure networks (water, electricity); - Physical imaging systems (acoustic and optical imaging); - Adaptive optics; - Energy conversion and distribution; - Biomedical systems, system biology. The programme brings together issues of physical modelling, experiment design, signal analysis and estimation, model-based control design and optimisation, hardware and software in a study of systems of high complexity and of different kinds, such as linear and nonlinear dynamics, hybrid and embedded systems, and ranging from small-scale micro-systems to large-scale industrial plants, structures, and networks.
Exit Qualifications	The MSc Systems and Control graduate possesses the following knowledge and skills: 1. Knowledge of engineering sciences (electrical engineering, mechanical engineering, applied physics, mathematics), in breadth and in depth, and the ability to apply this to systems and control engineering at an advanced level; 2. Scientific and technical knowledge of systems and control engineering, in breadth and in depth, and the skills to use this effectively. The discipline is mastered at different levels of abstraction, including a reflective understanding of its structure and its relationships with other fields, and to some extent this knowledge reaches the forefront of scientific or industrial research and development. Moreover, the knowledge possessed can form the basis for innovative contributions to the discipline in the form of new designs or the development of new knowledge; 3. Thorough knowledge of paradigms, methods, and tools, as well as the skills to apply that knowledge actively in analysis, modelling, simulation, design, and the conduct of research pertaining innovative, technologically dynamically systems, with an appreciation of different areas of application; 4. The ability to solve technological problems independently and in a systematic way, by means of problem analysis, formulating subproblems, and providing innovative technical solutions, including in new and unfamiliar situations. A professional attitude towards identifying and acquiring new expertise, towards monitoring and critically evaluating existing knowledge, towards planning and carrying out research, towards adapting to changing circumstances, and towards integrating new knowledge with an appreciation of its ambiguity, incompleteness, and limitations; 5. The ability to work both independently and in multidisciplinary teams, interacting effectively with specialists and taking initiatives where necessary; 6. The ability to communicate effectively about their work in the English language to both professionals and non-specialists, including the ability to make presentations and produce reports on, for example, solutions to problems, conclusions, knowledge, and considerations; 7. The ability to evaluate and assess the technological, ethical, and social impact of their work and to take responsibility with regard to sustainability, economy, and social welfare; 8. The willingness to maintain their professional competence independently through life-long learning.
Program Structure 1	FIRST YEAR (60 EC) The 1st year consists of: - Compulsory courses (35 EC), including the Integration project (5 EC); - Elective Systems and Control courses selected from a list provided (22 EC); - Social course: The student can choose between three courses, namely Philosophy of Engineering Science and Design, Ethics and Engineering, or Introduction to Safety Science (3 EC). SECOND YEAR (60 EC) The 2nd year consists of: - A choice between the following three options: 1) Joint Interdisciplinary Project (JIP - 15 EC) 2) Research project (10EC) and additional elective courses (5 ECTS); 3) Additional elective courses (15 ECTS). - Literature survey (Preparation of MSc thesis project) (10 ECTS). - MSc thesis project (35 ECTS).

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S&C Obligatory Courses (35 ECTS)	
Introduction 1	Curriculum The compulsory component of the curriculum, as shown in the table below, consists of a number of basic courses in key areas of systems and control: Control, Modelling, Nonlinear systems, Identification, Optimization and Signal processing. There is also a compulsory project component, namely the integration project. This project is performed in the laboratory, in which the knowledge acquired during the compulsory courses is applied to real-world situations.

SC42015	Control Theory	6
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	ME Department Delft Center for Systems and Control	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	Ir. S.C. Bregman	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis, and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the rotary pendulum, helicopter model, inverted pendulum on a cart, rotational pendulum, reaction wheel pendulum, and magnetic levitation system.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These courses cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, and system identification in open and closed loop. It is assumed that the students already know these concepts or are able to look them up in the literature. At the end of the course, the students should be able to: - Develop a mathematical model of a real-world dynamical system - Build a simulation model of the system in Matlab/Simulink - Identify the model parameters - Design a state estimator to recover unmeasured system states - Design controllers that regulate the derived system model - Integrate their acquired control engineering knowledge in laboratory setups	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. Due to the lab work that's needed to complete the project, it is required that you are enrolled on the Brightspace page and take part in the group/setup assignment procedure during the first week of the period. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC52035	
Department	ME Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After successfully completing this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 mandatory assignments Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.	
Assessment	Written exam (individual, fully closed-book, no calculators nor formulas sheets allowed, counts for 70% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 30% of the final marks). To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next.	
Permitted Materials during Tests	none	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SC42095	Digital Control	3
Responsible Instructor	A. Dabiri	
Instructor	M.A. Sharifi Kolarijani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	SC42015, SC42000 or similar. Knowledge of classical control techniques (systematic and realistic PID design, frequency domain approaches) as well as state space theory is required.	
Course Contents	Computer control. Sampling of continuous-time signals. The sampling theorem. Aliasing. Discrete-time systems. State-space systems in discrete-time. The z-transform. Selection of sampling-rate. Analysis of discrete-time systems. Stability. Controllability, reachability and observability. Disturbance models. Reduction of effects of disturbances. Design methods. Approximations of continuous design. Digital PID-controller. State-space design methods. Observers. Pole-placement. Optimal design methods. Linear Quadratic control. Prediction. LQG-control. Implementation aspects of digital controllers.	
Study Goals	The student must be able to: 1. describe the essential differences between continuous time and discrete-time control 2. transform a continuous time description of a system into a discrete-time description 3. calculate input-output responses for discrete-time systems 4. analyse the system characteristics of discrete-time systems 5. employ a pole-placement method on a discrete-time system 6. implement an observer to calculate the states of a discrete time system 7. apply optimal control on discrete-time systems 8. describe the functioning of the Kalman-filter as a dynamic observer	
Education Method	Lectures and computer exercises	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course.	
Literature and Study Materials	Course material: Lecture notes are made available on Brightspace Textbooks: K.J. Astrom, B. Wittenmark 'Computer-controlled Systems', Prentice Hall ,1997, 3rd edition. L. Keviczky et. al 'Control Engineering', Springer, 2019. L. Keviczky et. al 'Control Engineering: MATLAB Exercises', Springer, 2019. References from literature: B.C. Kuo 'Digital Control Systems', Tokyo, Holt-Saunders, 1980 G.F. Franklin, J.D. Powell 'Digital Control of Dynamic Systems', 1989, 2nd edition, Addison-Wesley	
Assessment	Project assignment	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: WB2305	
Department	ME Department Delft Center for Systems and Control	

SC42145	Robust Control	3
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	· Recap on background in linear systems theory and classical feedback control · Multivariable system control: Nyquist, interaction, decoupling · Directionality in multiloop control, gain and interaction measure · Stabilizing controllers and the concept of the generalized plant · Parametric uncertainty descriptions, approximations · The general framework of robust control · Robust stability analysis · Nominal and robust performance analysis · The H-infinity control problem · The structured singular value: Definition of mu · Mu synthesis, DK-iteration, role of uncertainty structure. · Design of robust controllers, choice of performance criterion and weights	
Study Goals	The student is able to reproduce theory and apply computational tools for robust controller analysis and synthesis. More specifically, the student must be able to: 1. substantiate relation between frequency-domain and state-space description of dynamical systems 2. define stability and performance for multivariable linear time-invariant systems 3. construct generalized plant for complex system interconnections 4. describe parametric and dynamic uncertainties 5. translate concrete controller synthesis problem into abstract framework of robust control 6. reproduce definition of the structured singular value 7. master application of structure singular value for robust stability and performance analysis 8. design robust controllers on the basis of the H-infinity control algorithm 9. apply controller-scalings iteration for robust controller synthesis	
Education Method	Lectures + Assisted lectures+Computer sessions	
Literature and Study Materials	S. Skogestad, I. Postlethwaite, Multivariable Feedback Control, 2nd Edition. John Wiley & Sons, 2005. References from literature: K. Zhou, J.C. Doyle, K. Glover, Robust and optimal control, Prentice Hall, 1996 D.-W. Gu, P.Hr. Petkov, M.M. Konstantinov, Robust Control Design with Matlab. Springer Verlag, London, UK, 2005	
Assessment	Controller design exercise (report and code)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC42010	
Department	ME Department Delft Center for Systems and Control	

SC42150	Statistical Signal Processing	3
Responsible Instructor	M. Khosravi	
Instructor	G. de Albuquerque Gleizer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	The course unit assumes that you are familiar with complex numbers, Z-transform, Fourier transform, basics of linear time-invariant systems, probability and statistics at the undergraduate level, and fundamentals of linear algebra. Also, programming skills in MATLAB/Python are necessary for the computer assignments.	
Course Contents	1. Theory of probability and point estimation 2. Stochastic signals 3. Linear filtering	
Study Goals	After the course, students should be able to: - Design and analyze parameter estimators - Analyze stochastic signals in both time and frequency domains. - Apply filtering techniques in the time and frequency domains. - Design linear time-invariant filters based on given frequency and time domain specs. - Estimate the information-carrying components from noisy measurement signals.	
Education Method	Lecture with tutorial consisting of theoretical and MATLAB/python exercises	
Assessment	Written exam(s) and homework. The final grade is a weighted average of the mark obtained on the examination(s) (80%) and that on the compulsory homework (MATLAB/python) assignments (20%).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42155	Modelling of Dynamical Systems	3
Responsible Instructor	Dr. R. Ferrari	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	The course will cover the basics of modeling of both physical systems (in the form of ordinary differential equations), computing and scheduling systems (as finite state machines), and combinations thereof (as hybrid models). Physical systems modeling. The course will cover the basics of Lagrangian modelling, I.e. employing energy balances and the Euler-Lagrange equations to arrive to closed form state-space dynamical equations for system analysis and control design. Bond-graph modelling will be covered as a powerful tool for modelling engineering systems, especially when different physical domains are involved. The basic ideas of model reduction, a technique to reduce the complexity of dynamical models, may also be covered in the course. The fundamental differences and connections between continuous vs discrete time modelling will be discussed. Similarly, frequency domain modelling is also discussed with practical motivations for frequency vs time domain modelling. Finite State Models. A brief introduction to models traditionally employed for computation, e.g. finite state machines, automata, and petri-nets, will be provided. Particular attention will be given to the applicability of these models to provide high-level dynamic descriptions of discrete-event systems and for behaviour specifications of designs. Hybrid systems. The course will finalize with a short introduction to hybrid systems through the formalisms of: Hybrid automata and jump-flow systems, and presenting a brief number of interesting subclasses of hybrid systems, e.g. PWA systems, and timed -automata.	
Study Goals	The purpose of the course is to introduce the students to basic concepts and results in physical modeling and the theory of nonlinear control systems. After successful completion of the course, the student is able to construct models of physical systems from the knowledge of the basic laws of physics (first principles) construct models of systems using the theory of bond graphs. construct models of systems using the Euler-Lagrange method (perform a model reduction for linear systems using a balanced realization) construct finite state models (automata, FSMs, Petri-Nets) of simple computing and scheduling systems analyze basic properties of finite state systems: blocking, determinism, and language expressed by a model. describe cyber-physical systems as hybrid dynamical models. select the appropriate modelling approach for a given problem.	
Education Method	Lectures 0/4/0/0	
Literature and Study Materials	Lecture notes and additional reader	
Assessment	Assignments (35%) and a mini-project (65%). The grade for the assignments will be the average grade of all the assignments. To pass the course, the final grade must be a 5.75 or higher. The grade for the mini-project is not allowed to be lower than a 5.0.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Embedded systems Group work Lineair Algebra Mathematics Matlab Mechatronics Modelling Numeric Methods Signals and Systems Stochastics	
Department	ME Department Delft Center for Systems and Control	

SC42165	Machine Learning for Systems and Control	3
Responsible Instructor	A. Dabiri	
Instructor	Dr.ing. R. van de Plas	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear algebra, basics of mathematical optimization, basics of probability theory	
Course Contents	The course will lay the methodological groundwork for machine learning in signals, systems, and control contexts and corresponding application domains. The course covers three modules: supervised learning, unsupervised learning, and reinforcement learning. Throughout the course, students will delve into topics such as linear regression, logistic regression, classification, neural networks, Bayesian supervised learning, and learn about model assessment techniques. In addition, the course covers clustering and matrix factorization, as well as fundamentals of reinforcement learning and its connection with dynamic programming. Furthermore, students will gain insight into model-free RL control algorithms like Sarsa and Q-learning, and approximation methods in value space.	
Study Goals	After this course, the student should be able to 1. Understand the theory of supervised learning 2. Apply state-of-the-art supervised learning algorithms for control applications 3. Understand basic unsupervised machine learning methods (e.g. clustering, principal component analysis) 4. Understand the link between reinforcement learning and dynamic programming 5. Apply RL algorithms like Q-learning or Sarsa for control applications	
Education Method	Lectures and computer exercises.	
Computer Use	Matlab , Python	
Literature and Study Materials	Course material: Lecture notes are made available on Brightspace Textbook: Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4. No. 4. New York: springer, 2006. Sutton, R. S., Barto, A. G., Reinforcement Learning: An Introduction. The MIT Press, 2018. Bertsekas D., Lessons from AlphaZero for Optimal, Model Predictive, and Adaptive Control, Athena Scientific, 2022 Li, S.E., Introduction to Reinforcement Learning. In: Reinforcement Learning for Sequential Decision and Optimal Control. Springer, Singapore, 2023. Hastie, T., Tibshirani, R., and Friedman, J., The Elements of Statistical Learning", 2nd Ed., Springer, New York, 2009.	
Assessment	Written exam and assignment.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

S&C Obligatory Social Courses: choose 1 out of 3 (3 ECTS)

WM0320TU	Ethics and Engineering	3
Module Manager	M. Sand	
Instructor	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%, individual grade), presentation and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course WM0329TU (6 ects).	
Elective	Yes	
Category	MSc niveau	

WM0349WB	Philosophy of Engineering Science and Design	3
Module Manager	Dr. J.M. Duran	
Instructor	Dr. J.M. Duran	
Responsible for assignments	Dr. J.M. Duran	
Co-responsible for assignments	Ir. S.J. Zwart	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Course contents: (1) The goals of science; the character and scope of scientific claims. (2) The goals of engineering design; the nature of technical artefacts; the value-neutrality of technology. (3) The scientific method and the validation of scientific claims. (4) Methods of engineering design; the character and scope of design claims; the decision-making aspect of design. (5) The development of science; the objectivity of science; the notion of scientific progress. (6) The development of technology; social determinism and technical determinism. (7) The place and role of values in science and in engineering design.	
Study Goals	This course aims first of all to support students in developing a reflective and critical attitude with regard to empirical research underlying engineering science and engineering design at an academic level. Additionally, and in support of this primary goal, it aims to make students acquainted with views on the nature and goals of science and engineering design and how in these activities facts and values both have their role to play and how they interact.	
Education Method	The course is taught in the form of seven weekly sessions which each consist of a plenary lecture of two hours and a seminar of two hours during which students articulate and discuss their answers to a number of questions in smaller groups. Each seminar will be prepared and chaired by a team of students. For this task general and specific instructions, recommendations and suggestions will be available.	
Assessment	Assessment is through (1) a final individual exam, (2) participation in the seminars and (3) performance in preparing and chairing a seminar session. The exam result forms 50 percent of the final grade, assessment of general seminar participation 25 percent and assessment of the preparing and chairing role also 25 percent. In the expectation that no COVID restrictions will be in place while the course is being offered, the exam will be a closed-book on-campus exam, which may be either hand-written or digital (ANS). Which of the two will be announced at the start of the course.	
Category	MSc level	

WM0801TU	Introduction to Safety Science	3
Module Manager	Dr. A.P. Afghari	
Instructor	Dr. A.P. Afghari	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. E. Papadimitriou	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Students wishing to graduate with a final project in the area of safety supervised by the Safety Science and Security Group should have followed this course or one of the other introductory courses run by the group (WM0808TU, WM0821TU or WM0822TU)	
Expected prior knowledge	Technical academic BSc and common (scientific) sense	
Summary	System Safety Engineering focuses on development of a safety oriented pattern of thinking and a holistic approach. The tools that will be gained in this course will be helpful in recognizing, understanding, and analyzing hazards; and assessing risks in contemporary complex systems. Areas included are (1) Hazard analysis and safety assessment, (2) Human reliability assessment, and (3) Safety management.	
Course Contents	Introduction to system safety engineering Safety performance measurement Hazard analysis Failure modes and effects analysis Preliminary hazard analysis System safety assessment Reliability block diagram Fault & event trees Markov chain analysis Bayesian network Consequence assessment Risk ranking Human reliability assessment Decision making under uncertainty	
Study Goals	The students will learn how to perform safety assessment, reduce risk within acceptable levels, manage risk, improve system safety, and make risk-informed decisions to benefit the organization and the community.	
Education Method	Plenary lectures (mandatory) Exercises (carried out in small groups of ~2 students)	
Literature and Study Materials	Clifton A. Ericson, Hazard Analysis Techniques for System Safety, John Wiley and Sons; ISBN: 978-0471720195. Harold E. Ronald and Brian Moriarty. (1990). System Safety Engineering and Management, John Wiley and Sons, 2nd Edition.	
Prerequisites	Technical background preferred but certainly not mandatory	
Assessment	Summative assessment: A final exam consisting of open-ended and multiple-choice questions; The exam is based on book, papers, exercises and lecture slides; The final exam is individual and accounts for 100% of the final grade for the course; A re-sit exam is possible if needed. Formative assessment: In-class quizzes, discussions and games; Sample questions for final exam.	
Enrolment / Application	Enrol through Brightspace. Minor students are automatically enrolled into the course.	
Remarks	This course can be expanded to a System Safety/Reliability project	
Elective	Yes	
Category	MSc level	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

S&C Elective Courses (min. 22 ECTS)

Introduction 1	
Program Structure 1	At least 21 ECTS should be taken from the below mentioned list of electives systems & control courses. At least 6 ECTS are free to choose of technical master courses. Within this 6 ECTS it is possible to select 6 points Internship (SC42115). In the second year the students can choose either to do a research project (10EC) or to take an additional 7 EC elective systems & control courses and 3 EC technical master courses (together 10 EC).

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory . Laplace transformation . Elementary closed loop systems . Transfer functions . Poles and zeros . First order systems . Second order systems . Pole placement for simple systems . Root locus method . Characteristic equation . Angle and magnitude conditions . State space formulation . Controllability, observability . Basic controllers: P,PI,PD,PID . Frequency response (Bode plots) 2. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 3. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 4. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 5. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 6. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 7. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 8. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance	

	- substantiate the function of a Flight Control System(FCS) in civil aviation. - apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4311	**Nonlinear and Adaptive Flight Control**	**4**
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I AE4301 AE4301P AE4312/AE4320	
Course Contents	Safety and survivability of modern civil and military aircraft when airframe and structural failures have occurred in flight have become a key issue in the aerospace industry nowadays. This course introduces nonlinear and adaptive control techniques to enhance reconfigurabilities of aircraft flight control systems. Advanced nonlinear and adaptive control techniques such as Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion (INDI), Adaptive Backstepping (ABS) and Incremental Backstepping are described together with applications to fault tolerant and reconfigurable aircraft flight control system designs. Unconventional yet available control effectors are used through control allocation algorithms to achieve required performance.	
Study Goals	Students will be able to acquire the skills in designing fault tolerant and reconfigurable flight control systems using Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion, Adaptive Backstepping (ABS) and Incremental Backstepping control techniques.	
Education Method	Lectures altered with assisted computer exercises	
Literature and Study Materials	Handouts of lecture slides (available on Brightspace) Recommended literature: J.-J.E. Slotine & Weiping Li, Applied non-linear control, Englewood Cliffs : Prentice-Hall, 1991, ISBN 0-13-040890-5 M. Krstic, et al, Nonlinear and adaptive contro design, John Wiley & Sons, Inc., ISBN 0-471-12732-9	
Assessment	Individual Take-home assignment	
Remarks	Hand-in exercise and a report of the design of an advanced controller. The report must be handed in at the request date given by the course instructor.	
Set-up	Course contents Introduction to advanced flight control Nonlinear aircraft flight control based on Nonlinear Dynamic Inversion Adaptive Nonlinear Dynamic Inversion Incremental Nonlinear Dynamic Inversion Aircraft fault tolerant flight control based on Adaptive and Incremental Nonlinear Dynamic Inversion Nonlinear control based on backstepping Nonlinear aircraft flight control based on backstepping Adaptive Backstepping Incremental Backstepping Aircraft fault tolerant flight control based on Adaptive and Incremental Backstepping Control allocation	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active at attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical sigularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hoever performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aeropace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course (optional). Presentation of a helicopter-related subject in a small Symposium (obligatory) One final take-home assignment at the end of the course (obligatory) +Oral Exam (obligatory)	
Special Information	A helicopter flight in SIMONA simulator may be given to the students following this course depending on the availability of SIMONA simulator.	
Set-up	Online education and Classical lecturing	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr.ir. C. de Wagter	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before/at the beginning of the course. Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight. The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo.	
Study Goals	(1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description of practical assignment: Student teams will develop an efficient, purely vision-based approach to navigate through an obstacle field autonomously. The algorithm must run onboard the Parrot Bebop Drone, a commercially available quadrotor with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist of 50% of the theoretical exam and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone in the given task, the approach as explained in a presentation, and the code produced by the team.	

AE4323	Real-time Distributed Flight and Space Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Instructor	Ing. E.H.H. Thung	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE1205 (Programming & Scientific Computing in Python, or equivalent introduction to programming) AE4322 (Piloted Flight Simulation)	
Course Contents	Lecture topics, not necessarily in chronological order 1) C++ introduction 2) Real-time software engineering 3) Delft University Environment for Communication and Activation (DUECA) 4) Modelling tools 5) Verification and Validation 6) Software project management Programming assignments, individually and in a team. The course is completed with a practical project in which students program, demonstrate and document a distributed simulation with an aerospace theme. Attendance in the project meetings is mandatory.	
Study Goals	1) The student can identify the software issues related to real-time, distributed simulation. 2) The student can actively apply the theory of rigid body dynamics in real-time simulation. 3) The student can combine the modelling of dynamical systems with numerical integration routines. 4) The student can use quaternions to represent rigid body attitudes. 5) The student can implement simulation models using different software languages and tools (C++, Simulink). 6) The student can create 3D models for computer visualisation and optimise these for real-time simulation. 7) The student can (in a team) implement a real-time simulation on a distributed platform, using advanced software tools.	
Education Method	Lectures, programming instruction, and programming assignments.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace.	
Assessment	Individual assignment during 2nd/3rd week and team project report. Attendance at the team project meetings in week 3 and later is mandatory.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr.ir. M.B. Zaayer	
Instructor	Prof.dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: describe the components and configurations of wind turbines describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. identify the drivers of dynamic behaviour and calculate their typical values describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Prof.dr. D.A. von Terzi	
Instructor	Dr. D. Zappalá	
Instructor	Dr.ir. M.B. Zaayer	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: Selecting approaches for design and analysis Collect information and data Use (simulation) programs, models and calculations Analyse and interpret results and draw conclusions about the design Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attented by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity AE4W31 Floating offshore wind energy ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AP3122	Advanced Optical Imaging	6
Responsible Instructor	Dr. K.S. Grußmayer	
Instructor	Prof.dr. S. Stallinga	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	BSc level courses on analysis, systems and signals, waves, and optics. Experience in programming with Python.	
Course Contents	We explain the basic properties of advanced optical imaging systems within the framework of Fourier optics and linear systems. We treat wave propagation, diffraction, phase transformation of lenses, modulation by apertures and objects. The spatial domain optical systems description will be transformed to the Fourier domain description of coherent and incoherent optical systems. Additionally, the course will treat various advanced imaging systems such as microscopes with different contrast mechanisms, optical coherence tomography, digital holography, and adaptive optics. We will show applications of microscopy in biomedical imaging. The course consists of lectures, pen and paper assignments, and (Python) programming assignments.	
Study Goals	The learning objectives of this course are: - Students can quantify the performance of optical imaging systems by applying Fourier optics methods - Students can theoretically describe the working of advanced optical imaging systems - Students can apply computational optics methods to model and analyze imaging concepts	
Education Method	Lectures, (pen and paper) homework, and computational optics assignments.	
Computer Use	Part of the course consists of making assignments in Python.	
Literature and Study Materials	Lecture notes and online material. The book: Introduction to Fourier Optics by J. Goodman is used as reference material.	
Assessment	Final written exam, computational optics assignments, and homework. The exam grade and the assignments grade both need a grade higher than 5.0. Homework is assessed by pass-fail. A maximum of one point bonus is provided on the end grade according to the ratio #passes/6. The final grade is computed with the following formula: Final grade = 0.6*(Exam grade) + 0.4*(Average assignment grade) + (Homework). The validity of result of the computational optics assignments and homework is restricted to one academic year.	
Permitted Materials during Tests	Handwritten notes page.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/education/admission-and-application/elective-course#c1323630	

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Prof.dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 7 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3382	Advanced Photonics	6
Responsible Instructor	Dr. O. El Gawhary	
Instructor	Prof.dr. H.P. Urbach	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Electromagnetism bachelor level, optics course bachelor level. Some knowledge about Fourier transforms and mathematics for physicists at the bachelor level will be useful.	
Course Contents	The course consists of a basic part, which provides all the background needed to study the advanced applied topics, and of a second part where we discuss advanced subjects, which are barely encountered in any textbook but that are of vital importance for any modern applications of optics. In the first part of the course we discuss some of the foundations of optics, embedded in the more general context of electromagnetic theory. Here below some more detail on the content. 1. The basic part provides rigorous (namely at a higher and deeper level than usually treated in other courses and books) treatments of the basic theoretical aspects of optics. This material can be found scattered in different references, but we also provide lecture notes where the theory is described. The material contained in those lecture notes is much more extensive than we can cover in the course. During the first lectures, we start with the electromagnetic theory of optics. We will focus on how to formulate and solve scattering problems in optics using the Lippmann-Schwinger equation. We will discuss how to solve that fundamental equation using perturbation methods (the Born series) and we will explain why this method seems to fail in practice and what can be done about it. We will recall diffraction theory and we will make use of the stationary phase method to approximate rapidly oscillating integrals, which are often encountered when dealing with propagation problems. The plane wave expansion method to propagate optical fields through homogeneous space is derived. That method is important as it will give us the chance to understand the fundamental limit that exists for the amount of spatial information about an object that can be transferred by any monochromatic light wave (although to really understand why such a resolution limit exists we will need to discuss on inverse problems, ill-posedness, regularization techniques etc, which will do as well). Equipped with this knowledge, you will be able to also understand the true origin of classical resolution limit of optical imaging, which is instrumental to trying to figure out ways to overcome that limit. Then, depending on the applied topics which we will choose to study in the second half of the course, we discuss one or a few topics out of: a) optical beams; b) Parameter estimation and inference; c) partial coherent light and statistical optics; d) holography and phase sensitive imaging; Which of these subjects is covered depends, among other things, on the knowledge and interest of the students taking the course how long and how extensive the basic theory will be treated. 2. In the second half of the course we will apply the theory to study advanced applications. Examples are new (proposed) methods to achieve super-resolution by using meta-materials, hyperbolic materials ad multiple scattering. Plasmonic waves and their role in achieving high sensitivity will be treated as well as the concepts of a perfect and super lens are explained. Furthermore, regularization methods in parameters retrieval and inverse scattering problems are discussed. Also new ways of imaging, such as lensless imaging using phase retrieval techniques and ghost imaging are discussed. Optical beams with special properties (non-diffracting beams, beams carrying an orbital angular momentum, phase singularities) may be also discussed. More advanced topics, such as non-trivial symmetries and conservation laws in optics can also be discussed. In many cases, the topics are treated by studying scientific papers together which will be made available. The particular choice of topics will also depend on the preference of the students. Finally, the course is made of tutorials (typically 1 hour a week), exercise sessions and normal lectures. In the tutorial we will show the detailed solution to problems while the exercise sessions are mostly meant to gain practical experiences in solving problems	
Study Goals	1. Students acquire an in depth knowledge about the basic theory of optics. The topics are discussed starting from few common-sense assumptions and starting from those, the theory develops as a natural sequence of logical steps. 2. Students learn the hierarchy of optical models (electromagnetic optics, diffraction optics, geometrical optics) and the the modelling of coherence of light and know when these models can be applied. 3. Students practice reading scientific papers in the field of optics and develop a critical attitude.	
Education Method	oral lectures, tutorials and self-study. Lectures and tutorials are given in a blackboard-like approach (slides are used only very occasionally, to show some figure or the result of a simulation). All material treated during the lectures is then made available to the students in Brightspace.	
Literature and Study Materials	All lecture materials are handed out and will be put on Brightspace. Most of the material is a selected from lecture notes which are also available on Brightspace.	
Assessment	Written or oral exam (depending on the number of students).	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/education/admission-and-application/elective-course#c1323630	
Elective	Yes	

AP3391	Geometrical Optics	6
Responsible Instructor	Dr. F. Bociort	
Instructor	Dr. B. Kruizinga	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	This course consists of two parts: the larger part is about the theoretical basis of geometrical optics, the second part is an introduction to optical system design. The topics are: - Fundamentals of geometrical optics (geometrical optics as a limiting case of wave optics, the eikonal function, rays and wave fronts, ray paths in inhomogeneous media) - Ray tracing (Snells law in vector form, formalism for reflection, refraction and transfer, ray failure, aspherical surfaces) - The paraxial approximation (paraxial and finite rays, matrix formalism, characteristics of ideal imaging, principal planes, telescopic systems, aperture and field stops, pupils, vignetting, marginal and chief rays, Lagrange invariant, F number, telecentric systems) - Aberrations (transverse ray aberration, wave front aberration and the relationship between them, power series expansions for optical systems with or without rotational symmetry, rotationally invariant combinations of ray parameters, defocusing, Seidel aberrations. Aberration balancing, caustic, Design aspects: situations when some aberrations are more important than others, aplanatic surfaces, ideal placement of aspheric surfaces.) - Optical design software, local and global optimization of optical systems.	
Study Goals	- Mastery of the concepts, theories and methods listed above at an advanced academic level. - Development of the ability to perform simple lens design tasks with state-of-the-art lens design software.	
Education Method	Oral lectures and independent work with lens design software. Because of the work with software, late enrollments in this course (more than 10 days after the 1st lecture) are not accepted.	
Literature and Study Materials	1. J. Braat, Diktaat Geometrische Optica , TU Delft 1991 (in Dutch; English-speaking students should use Born and Wolf (Ref 5) instead); 2. J. Braat, Paraxial Optics Handout (on Brightspace); 3. W.T. Welford, Aberrations of Optical Systems, Adam Hilger, 1986 (or the earlier version Aberrations of the Symmetrical Optical System, 1974); 4. F. Bociort, Optimization of optical systems (can be found on Brightspace). Supplementary reading (not required for the exam, just if you want extra depth on some subject): 5. M. Born and E. Wolf, Principles of Optics; 6. R.R. Shannon, The Art and Science of Optical Design, Cambridge University Press, 1997; 7. D. Sinclair, Optical Design Software, Handbook of Optics, Chapter 34.	
Assessment	The final mark will be for 70% the mark for the theoretical part (oral exam) and for 30% the mark of the design part (submitted solutions to exercises). To pass the exam, for both parts the minimal mark is 5.0.	

AP3401	Introduction to Charged Particle Optics	6
Responsible Instructor	Dr. A. Mohammadi Gheidari	
Instructor	Prof.dr.ir. J.P. Hoogenboom	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Geometrical optics: focal point, thick lens model, matrix description, phase space, Liouville, aberrations; Electricity and magnetism, Maxwell's equations, Laplace equation, flux lines, equipotential planes; Fourier analysis, numerical methods, series expansions.	
Course Contents	Electron and ion beam systems are pivotal for modern-day science and technology. They provide us with the capability to see, manipulate, and make things at the smallest length scales, down to a nanometer or even almost atomic resolution. In this course you will learn the basics about the design of these instruments, and about how to do optics with beams of charged particles. You will learn how sources, lenses, aberration correctors for electrons and ions work and how they can be optimized together for best performance. Topics treated include: Electron sources, electrostatic lenses, magnetic lenses, multipole elements, application and simulation of multipole elements, off-axes aberrations, Coulomb interactions, electron and ion beam instruments (microscopes and lithography tools), MEMS electron optics.	
Study Goals	After successful completion of the course, the student is able to: - Understand and explain the principles of charged particle optics - Understand and explain the basic working principles of electron and ion beam instrumentation - Apply the principles of charged particle optics to design electron optical parts of electron and ion beam instrumentation - Summarize key concepts of charged particle optics in a written report - Collaborate in a team of 2 or 3 people to create solutions for electron optical problems and create a written report on the solutions which contains a critical reflection on the own work - Identify applications of charged particle optics in science and industry	
Education Method	Two hours of lecture per week followed by self-study and assignment work and reporting in teams of 2 or 3 students. Each lecture consists of one hour introduction of the new study and assignment subject and one hour of discussion of the reports and solutions of the previous assignment. In total there are 12 assignments. All assignments are graded, the 10 highest count towards the final grade. There is no exam and no resit opportunity within the academic year.	
Literature and Study Materials	course book and material on Brightspace.	
Reader	Reader digitally available, a hard copy can be obtained via the instructors (by payment of printing costs)	
Assessment	Assignment reports (12 in total) are graded, the 10 highest grades count towards the final course grade. There is no exam and no resit opportunity with in the academic year.	
Elective	Yes	
Tags	Elektricity & Magnetism Physics Technology	
Studyload/Week	8 hours per week	

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course covers the following topics: - basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and terminology of real-time systems 2. Construct task schedules using different scheduling policies under a given set of realistic system constraints 3. Analyze the timing behavior of a system for a given system model and scheduling policy 4. Discuss advantages and disadvantages of different scheduling policies for a given platform or system 5. Discuss the effect of hardware and software interferences on the timing behavior of a given system 6. Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system 7. Derive (reverse engineer) the system specification from a given implementation (in the lab) 8. Evaluate the scheduling overheads of a given implementation (in the lab) 9. Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Individual Assignment: lab work (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 6 (5.8 and higher). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CIEQ6002	Transport Modelling and Analysis	5
Responsible Instructor	Dr. S. Sharif Azadeh	
Instructor	Dr.ir. A.J. Pel	
Instructor	Prof.dr. O. Cats	
Instructor	Dr. S. Sharif Azadeh	
Contact Hours / Week x/x/x/x	x/5/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Four-step transport models, output in terms of travel time, congestion, flows and modelling of impact on accessibility, traffic safety and environment. Demand modelling: trip generation, distribution and modal split using methods such as gravity models and OD estimation. Assignment problems part I (Uncongested networks and stochastic assignment, Congested networks, equilibrium assignment). Assignment problems part II (Public transport assignment System optimal assignment and dynamic assignment). Introduction to Markov decision processes and reinforcement learning. Projects topics and results associated with these models will be shared with students as future impactful research opportunities. Location planning part I (Facility location problem (service point). Application: location of charging stations for EVs.) Location planning part II (Facility location problem (transhipment hubs). Application: multimodal mobility service, DRT integrated in Fixed Line and Schedule systems).	
Study Goals	On completion of this Course, the student will be able to: TBU2-LO1: Describe and apply four-step transport models both for mobility and freight (TTE-TB-LO3, TTE-TB-LO4). TBU2-LO2: Describe and apply the key performance measurements of a transport system and their impacts (TTE-TB-LO2, TTE-TB-LO5). TBU2-LO3: Describe and estimate travel demand models at aggregated and disaggregated levels (TTE-TB-LO3, TTE-TB-LO4). 331 TBU2-LO4: Describe, compare and apply assignment models combining travel demand and transport supply models for road and public transport (TTE-TB-LO3, TTE-TB-LO4). TBU2-LO5: Formulate a location planning problem (service point and transhipment hubs) as a mixed integer linear model and solve using a branch and bound (cut) algorithm ((TTE-TB-LO3, TTE-TB-LO4).	
Education Method	Lectures, in class small groups practice problems, in-class discussions	
Assessment	Written exam (100%)	
Expected prior Knowledge	Technical Bc. esneciallv with resnect to calculus. statistics. the concents of mathematical modelling and algorithms Knowledge on transport systems in general is recommended.	
Academic Skills	Lecture notes and scientific articles provided by the teachers and are made available through Brightspace.	
Literature & Study Materials	Course slides and provided reading material	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Written exam (100%)	
Permitted Materials during Exam	Non-programmable calculator	
Collegerama	No	

CIEQ6003	Traffic Modelling and Management	6
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Instructor	Dr.ir. A. Hegyi	
Instructor	Dr.ir. A.M. Salomons	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. H. Farah	
Contact Hours / Week x/x/x/x	x/x/6/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Roadway capacity and cumulative curves Intersection control Car-following behaviour, micro-macro relations, trajectories Railway capacity Timetable stability and max-plus algebra Speed contour plots & capacity drop Fundamental diagram & shock waves Traffic flow properties for buses, trams and metro, incl. macroscopic fundamental diagrams and dwell time functions. Transit signal priority. Pedestrians: expand traffic flow theory to two dimensional planes Safety, environment; control objectives Systems and control Network management - hierarchical and multi-level (Application of) traffic simulation tools (reliability, seeds, ...)	
Study Goals	On completion of this course, the student will be able to: TBU3-LO1 Describe traffic flows in space and time at microscopic and macroscopic level (TTE-TB-LO1) TBU3-L02 Discuss and apply theories and models at microscopic and macroscopic level for different transport modes (TTE-TB-LO3). TBU3-ILO3 Develop and assess control strategies at local and network level (TTE-TB-LO2). TBU3-ILO4 Set-up, perform, and report a traffic flow simulation experiment (TTE-TB-LO4).	
Education Method	Lectures, feedback on group work	
Assessment	Written exam (100%)	
Elective	Yes	
Tags	Analysis Modelling Transport & Logistics Transport phenomena	
Expected prior Knowledge	Calculus; basic insights into modelling and insights into simulation setup.	
Academic Skills	Critical thinking	
Literature & Study Materials	Lecture notes provided by the teachers	
Judgement	100% exam For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Self-made notes (limited in size; see brightspace)	
Collegerama	No	

CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4
Responsible Instructor	Dr.ir. I. Martínez Josemaría	
Instructor	Dr.ir. S.C. Calvert	
Instructor	Dr. M.P. Hagenzieker	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr.ir. M. Rinaldi	
Instructor	Dr.ir. I. Martínez Josemaría	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Introduction, classification and functional description of intelligent vehicles; New mobility services and system level impacts; Motion planing, decision and control methods for intelligent vehicle systems; Technologies for intelligent vehicles, sensors, communication, and actuators; Behavioural adaptation to intelligent vehicles; Impact assessment of intelligent vehicles on traffic safety; Impact assessment of intelligent vehicles on traffic efficiency; Simulation experimental research of impacts of intelligent vehicles on traffic flow using traffic simulation	
Study Goals	On completion of this course, the student will be able to: CIEQ6223-LO1. Analyze the essential components and classifications of intelligent vehicles. CIEQ6223-LO2. Propose the necessary sensing and communication components and objectives for your intelligent vehicle design. CIEQ6223-LO3. Formulate the tactical and/or operational level decision-making systems of your intelligent vehicle system. CIEQ6223-LO4. Justify behavioural adaptation effects of intelligent vehicles. CIEQ6223-LO5: Evaluate the system performance of new transport modes that emerge due to vehicle automation. CIEQ6223-LO6. Implement an intelligent vehicle control strategy for limited operational design domain navigation tasks using python. CIEQ6223-LO7. Assess the traffic safety and traffic flow impacts of intelligent vehicle systems through traffic simulations.	
Education Method	Lectures (2x 2 hours per week)	
Assessment	100% group assignment.	
Expected prior Knowledge	The course is closely related to CIEM6220-U4 Traffic Flow Modelling and Control and CIEM6220-U1 Traffic Safety.	
Academic Skills	In addition to the technical materials, students will also practice literature research, presenting, writing skills, giving feedback, and working in teams.	
Literature & Study Materials	Recommended papers and slides	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	None	
Collegerama	No	

CIEQ6224	Urban and motorway Traffic Flow Modelling and Control	5
Responsible Instructor	Dr.ir. A. Hegyi	
Instructor	Dr.ir. A.M. Salomons	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. Y. Yuan	
Contact Hours / Week x/x/x/x	x/x/x/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Modelling: Macroscopic traffic flow models, node models Shock wave theory - moving bottlenecks Network fundamental diagram Control: Traffic state estimation Traffic control design methodology Applications: Advanced vehicle-actuated signal control Urban network control Freeway traffic control (local and coordinated ramp metering, route guidance, variable speed limits)	
Study Goals	On completion of this course, the student will be able to: B2U4-LO1. Describe and apply macroscopic traffic flow models, such as the Cell Transmission Model, node models, to determine the evolution of traffic over time; B2U4-LO2. Discuss the shape of the macroscopic fundamental diagram and explain perimeter control; B2U4-LO3. Apply and explain Kalman filtering for traffic state estimation problems; B2U4-LO4. Explain the different purposes and designs of (1) coordinated ramp metering, (2) route guidance, (3) variable speed limit systems. Explain and apply the discussed designs for such systems to a traffic situation. B2U4-LO5. Design and analyse a vehicle-actuated intersection controller. B2U4-LO6. Describe and discuss the main features of network-oriented urban signal control approaches. B2U4-LO7. Explain the steps of the design methodology for traffic control systems. Apply the steps to design problems.	
Education Method	Lectures (2x2 hours per week) Practical by supervised work on exercises for assignments (1x4 hours per week)	
Assessment	Written exam (100%)	
Expected prior Knowledge	Road traffic flow modeling and control related parts of CIEQ6003.	
Academic Skills	Application of theory to new situations, conceptual thinking, critical thinking, problem-based learning and solving.	
Literature & Study Materials	Lecture notes provided by the lecturers.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Equation sheet will be handed out at (and uploaded to Brighspace). Simple calculator. One A4 sheet (two-sided) with notes in own(!) handwriting, max 30 lines, 80 characters per page.	
Collegerama	No	

CIEQ6231	Public Transport System and Supply Planning and Operations	5
Responsible Instructor	Dr.ir. N. van Oort	
Instructor	Prof.dr. O. Cats	
Contact Hours / Week x/x/x/x	x/x/x/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Service analysis Transit data sources and performance analytics Data fusion and analysis Technology design and properties (BRT, light rail,...) Access and egress, Cycling and transit integration, Shared micromobility Emerging modes and concepts (autonomous shuttles, MaaS, Hubs) Service design Frequency setting Fleet size determination Resource allocation and vehicle scheduling Control Reliability analysis Real-time control measures Societal aspects of the entire public transport planning process and operations are discussed. Also, the (expected) impacts of new technologies and emerging modes on sustainability and equity will be evaluated. The part of the unit dealing with first and last-mile solutions will discuss bicycle-related and micromobility solutions.	
Study Goals	LO1. Explain and relate the functions of strategic, tactical and real-time operation and control of public transport systems, first and last-mile solutions and emerging modes and concepts. LO2. Devise a data collection and analysis scheme directed at gaining knowledge on travel patterns and system performance of public transport. LO3. Compute and analyse the capacity and reliability of public transport services. LO4. Construct timetables and vehicle schedules for public transport services. LO5. Identify and evaluate potential control measures and to learn when and where to apply them, considering trade-offs of impacts on passengers, crew, vehicles and costs.	
Education Method	On-campus lectures from instructors Multiple industry lectures from operators and consultancy companies Active learning activities such as a data challenge, a serious game and small class exercises	
Assessment	Written exam (100%)	
Literature & Study Materials	Lecture slides and multiple papers per week provided during the course	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEQ6232	Public Transport Demand and Network Planning and Operations	5

Responsible Instructor Prof.dr. O. Cats

Instructor Dr.ir. N. van Oort

Contact Hours / Week x/x/x/x x/x/x/4

Education Period 4

Start Education 4

Exam Period 4
5

Course Language English

Course Contents

Modelling PT planning problems. The course consists of three blocks: network and demand analysis, network assignment and design, and service design.

The course covers the following topics:
- Planning process
- Network analysis
- Demand and behavioral modelling
- Network assignment
- Network design
- System resilience
- Cost-benefit analysis
- Data and modelling in planning practices
- Demand and supply real-time management
- Mobility-on-Demand
- Service optimisation in operations and planning practices
- Long-distance travel

The course includes lectures, lab tutorials and talks highlighting insights from practice.

Societal aspects of the entire public transport planning process are discussed. The unit will discuss the relations between public transport service design and societal/urban development goals such as accessibility, coverage, sustainability, equity and resilience. Attention is also given to communicating modelling and analysis results to decision makers and implications of mobility service developments for different user groups.

This unit is the official substitute of the course CIE4811 "Planning and Operations of Public Transport Systems".

Study Goals

LO1. Analyse key dependencies among design variables, parameters, and criteria in public transport service planning and management and their dynamics.

LO2. Design and examine public transport networks while considering coverage, accessibility, travel and operational costs.

LO3. Characterize the structure of public transport networks by applying graph theory techniques.

LO4. Compare alternative network models, simulation, optimization and choice modelling techniques in the realm of public transport, and their key notions, strengths, and weaknesses.

LO5. Select the modelling approach adequate for a given application concerning network modelling, optimization and assignment models, and interpret the results of these models

Education Method

The course involves lectures, in-class discussions, feedback sessions, pre and post-lecture material such as reading scientific articles and watching videos.

Assessment

Group assignment including an individual component: 100%.

Tags

Algorithmics
Analysis
Business
Modelling
Programming
Sustainability
Transport & Logistics
Transport phenomena

Expected prior Knowledge

This courses provides knowledge on planning and operations of public transport systems and does not have specific prior knowledge requirements. Notions related to network science, transport fundamentals and mathematical programming will be used in several classes.

Academic Skills

The main academic transferable skills that will be developed in this unit are critical
thinking, analytical thinking and problem solving, specifically applied to the planning and operations of public transport networks and demand analysis.

Students are expected to exercise the following academic skills:
- Problem formulation
- Data analysis
- Statistical interpretation
- Comparing alternatives quantitatively and qualitatively
- Working in diverse groups
- Critically reflecting on their own work
- Report structuring and writing
- Reading and summarizing scientific papers

The main personal skills that the student will develop are organizational, team-building, and interpersonal and presentation abilities by means of concrete cooperation with colleagues in assignments having the scope of emulating projects and interactions in a professional environment.

Literature & Study Materials

Lecture slides and pre- and post-lecture material including 3-5 reference
papers are made available through Brightspace.

Judgement

See Assessment

For more information on grading, see article 14 in the Rules and Guidelines (RGBE):
https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

Permitted Materials during Exam Not applicable (no exam)

Collegerama	No

CIEQ6233	**Railway Operations and Control**	**5**
Responsible Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Dr. D.M. van de Velde	
Contact Hours / Week x/x/x/x	x/x/x/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Railway safety and signalling Railway safety principles Train separation, Interlocking, Automatic train protection, Level crossings Blocking time theory Railway capacity assessment Next generation signalling (ERTMS, CBTC, Virtual Coupling) Migration to ERTMS Railway automation Automatic Train Operation Grades of Automation ATO-over-ETCS Railway traffic control Metro control Railway governance and production chains Railway governance Railway reform Open access Railway production chains Short-term, Mid-term and Long-term coordination This unit will underline the impact of railway signalling and automation on railway performance and public safety, making the students aware of the responsibilities and ethical implications related to the introduction of fully automated railway operations. Roles and responsibilities of the different railway system components and stakeholders are introduced, alongside with potential conflicts of interest in the railway business. Regulations and policies over the entire railway production chain are discussed in line with international objectives on sustainable mobility.	
Study Goals	LO1: Assess railway safety and signalling systems LO2: Assess railway automation LO3: Analyse railway production chains and coordination misalignments LO4: Evaluate railway governance and main policy options LO5: Evaluate ERTMS migration issues	
Education Method	This course uses a mixture of lectures, reading, guest lectures, and formative exercises.	
Assessment	Written exam (100%)	
Expected prior Knowledge	None	
Academic Skills	Modelling, design, critical thinking, reasoning	
Literature & Study Materials	J. Pachl, Railway Signalling Principles, 3rd ed., 2024. And a set of papers and book chapters made available on BrightSpace. Each lecture will require reading one (short) paper/book chapter.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Simple calculator	
Collegerama	No	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4291	Control of Electrical Drives	5
Responsible Instructor	Dr. J. Dong	
Contact Hours / Week x/x/x/x	practicum 0/0/0/4 vanaf week 2, instructie 0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Course Contents	- Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control. - Practical design of an electrical drive system.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system. - Design the controllers in electrical drive.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Literature and Study Materials	- Rik De Doncker, Duco W.J. Pulle, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Reader	"Reader Control of Electrical Drives ET4291 Version maart 2009"	
Prerequisites	Electrical Machines and drives (ET4117) Electronic Power Conversion (ET4119) AC Machines (ET4121)	
Assessment	Individual assignment and oral exam In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ME41006	Musculoskeletal Modeling and Simulation	4
Responsible Instructor	A. Seth	
Instructor	Ir. J.D.C. Groenhuis	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Human and animal movement is often smooth, powerful, and graceful. Animal movement is generated through a cascade of nervous, muscular, and skeletal system dynamics. Even the simple tasks of walking and running require the coordination of hundreds of muscles that pull on bones that move our limbs and generate forces on the environment so we can move. When these systems are disrupted through disease and injury, they result in movement disorders that can be debilitating. In this course you will learn to construct and simulate musculoskeletal models that generate movement. You will learn how to apply models to improve measurements of movement and to predict behavior, thus extracting more insights about human (and animal) movement than from observations alone. We will start with the constituent components of bones, joints, muscles, and tendons and their dynamics, as the building blocks of musculoskeletal systems. You will learn inverse techniques to extract unmeasured quantities, like joint kinematics, work, and power from observations and to use forward dynamic simulations to predict behaviors and answer what if? type questions. You will apply these models and techniques to answer a research question of your own formulation	
Study Goals	The goal of this course is to teach you how to use state-of-the-art computational tools to model and simulate the human musculoskeletal system. Learning will be guided by lectures, reading articles, practical labs and a final project. Upon successfully completing this course, you will be able to: 1. Model Develop and analyze mathematical models of the human musculoskeletal system using OpenSim and MATLAB. 2. Simulate Create inverse and forward dynamic simulations of human movement. Perform inverse dynamic simulations and optimizations using experimental data to compute quantities that cannot be easily measured (e.g., muscle forces). Perform forward dynamic simulations to reproduce observed movements and predict new movements. 3. Analyze Use musculoskeletal simulation to answer questions about human movement (e.g., how does a muscular impairment affect muscle coordination?), and validate results against experimental data. 4. Critique Read journal papers in the field; establish the main contributions and limitations of the work. Summarize your work using a thoughtful structure and relate your work to the broader field. You will also review the work of your peers and provide feedback on clarity, correctness and impact. 5. Formulate a research question and write a corresponding proposal.	
Education Method	Flipped Classroom method. Lecture slides, reading materials, and assignment descriptions provided through Brightspace. Class time is used to review reading materials and answer questions on concepts and practical labs. The bulk of the work is performing challenge based labs and a final project of your choice.	
Assessment	60% Three practical computer lab assignments (20% each) 40% Final project (proposal, presentation, abstract, reproducibility, reviews) Pass 6/10 grade total	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME41056	Multibody Dynamics	5
Responsible Instructor	J.K. Moore	
Instructor	W. Wolfslag	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Vector calculus, Physics, Linear algebra (e.g. Wiskunde 2 Lineaire Algebra WBMT1051), Differential equations (e.g. Wiskunde 3 Differentiaal vergelijkingen WBMT2048 T2), Numerical methods (e.g. Wiskunde 4 Numerieke Wiskunde WBMT2049 T2), Statics (e.g. Statica WB1630-20), Dynamics (e.g. Introductory Dynamics WB1135 and/or Rigid Body Dynamics WB2630), Scientific computing (Python, Matlab, Julia, etc.)	
Course Contents	Multibody systems are mechanical systems made up of multiple interconnected rigid bodies. In this course, you will learn a systematic approach for generating, solving, and integrating equations of motion used to model multibody systems. Topics covered are: - Three dimensional kinematics and use of Euler angles and Tait-Bryan angles for describing orientation. - Equations of motion for planar and three-dimensional systems, free body diagrams, constraint equations and constraint forces, uniqueness of the solution. - Systematic approaches for generating equations of motion, e.g. Newton-Euler, Lagrange, coordinate transforms (TMT), Kane, etc. - Relationship of maximal coordinate equations of motion to generalized coordinate equations of motion. - Configuration (holonomic) constraints, e.g. closed kinematic loops, and motion (nonholonomic) constraints, e.g. rolling without slipping, and their relationship to degrees of freedom. - Symbolic and numeric computational methods applicable to generating and solving equations of motion. - Numerical integration of coupled differential and differential algebraic equations with careful attention to stability, accuracy, and minimization of integration drift. - Visualization and analysis of two- and three-dimensional motion.	
Study Goals	The student is able to find the motions of linked rigid body systems in two and three dimensions including systems with various kinematic constraints such as sliding, hinges, rolling, and closed kinematic loops. More specifically, the student must able to: 1. derive the equations of motion for a simple planar system, draw free body diagrams, set up constraint equations, and demonstrate the uniqueness of the solution 2. derive the equations of motion for a system of interconnected rigid bodies by means of a systematic approach 3. derive the equations of motion in terms of generalized independent coordinates 4. apply the techniques from above to systems having holonomic and nonholonomic constraints 5. perform various numerical integration schemes on the equations of motion, and predict the stability and accuracy of the applied methods 6. perform numerical integration on a coupled system of differential and algebraic equations (DAE's) with minimal constraint drift 7. derive the equations of motion for a general rigid three-dimensional body system bound by constraints 8. identify the need to describe orientation in 3D space by means of: Euler angles and Tait-Bryan angles and derive the angular velocity and accelerations in terms of these parameters and their time derivatives 9. create accurate, efficient, and documented computer programs that construct and simulate multibody systems 10. analyze and interpret the resulting motion from a multibody simulation	
Education Method	Lectures, homework, and work sessions	
Computer Use	Laptop needed in work sessions	
Assessment	Weekly assignments and an exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME45155	Computational Fluid Dynamics for Engineers	5
Responsible Instructor	Dr.ir. M.J.B.M. Pourquie	
Instructor	Prof.dr. R. Pecnik	
Instructor	Dr. P. Simões Costa	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	WB2542 (thermofluids), ME45042 or ME45043 (advanced fluid dynamics), WBMT2049 T2 and T3 or similar (basic numerical analysis), some elementary programming skill (python, matlab or any other)	
Course Contents	This is a course on computational fluid dynamics related to process and energy systems. Some elementary knowledge of numerical methods (discretisation of a derivative, numerical solution of ordinary differential equations) is assumed. See expected prior knowledge. In general the course will cover: (1) general conservation equations for mass, heat and momentum transport (Navier-Stokes equations) (2) numerical solution of multi-dimensional, stationary and time-dependent problems as encountered in low-speed CFD applications; (2.1) numerical schemes for temporal discretization; (2.2) numerical schemes for spatial discretization; (2.3) imposition boundary conditions (3) turbulence modeling (4) wave equation (5) numerical methods for high-speed (compressible) flows (6) high performance computing	
Study Goals	After learning the content of the course the student will have the following capabilities: (1) Describe the two most popular methods in commercial CFD, finite differences and finite volumes (2) solve simple demonstrative problems in fluid flow and heat transfer by programming them in python or matlab, using finite differences and finite volumes (3) recognize the effects of numerical methods on the solution, such as numerical diffusion and numerical dispersion and to explain how to make these effects smaller (4) recognize numerical instability, to list several ways to avoid it and to analyze stability of simple methods analytically (5) solve fluid flow and heat transfer problems with the commercial CFD package Fluent or the open source CFD package OpenFOAM, which includes the following: set up a calculation, generate a simple grid, set appropriate boundary conditions, choose suitable discretisation, interpret and validate the results. Alternatively, in some cases, instead of using a commercial package, students can decide to write and validate their own code which they will use for calculating a fluid mechanics problem.	
Education Method	Lectures, practical exercises (2x2 hours per week)	
Literature and Study Materials	Course material: (1) Sheets/handouts (2) J.H. Ferziger and M. Peric, Computational methods for Fluid Dynamics, Springer Verlag. (3) P. Moin, Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2001. References from literature: (1) C. Hirsch, Numerical computation of internal and external flows, Volume I Fundamentals of numerical discretization, Volume II Computational methods for inviscid and viscous flows, Chicester, Wiley & Sons, 1988, 1990 (2) C.A.J. Fletcher, Computational techniques for Fluid Dynamics, Volume I Fundamental and general techniques, Volume II Specific techniques for different flow categories, Berlin, Springer, 2-nd ed. 1991. (3) H.K. Versteegh, W. Malalasekara, An introduction of computational fluid dynamics. The finite volume method. Second edition. Pearson Education.	
Assessment	Written exam + assignment with report; written exam may be online type depending on circumstances	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME46010	Intro to Nanoscience and Technology	3

Responsible Instructor
Dr. S. Caneva

Instructor
Prof.dr. U. Staufer

Contact Hours / Week x/x/x/x
0/0/4/0

Education Period
3

Start Education
3

Exam Period
3
4

Course Language
English

Required for
Micro and Nano Engineering within the PME track

Expected prior knowledge
ME46005 or equivalent course on classic physics (mechanics, electrodynamics, waves)

Summary
Introduction into concepts, methods, instruments, and processes used in nanotechnology

Course Contents
Nanoscience emerged from the analysis of basic physical, chemical and biological phenomena at the atomic to sub-micrometer scale range.
This course first establishes the basic fundamental knowledge and terminology used in nanoscience and nanotechnology. We will then cover advances in new materials, processes, devices and instruments to generate, manipulate and image nanostructures. The following topics will be treated:

The basics:
- Photon: the quantization of light
- de Broglie: Electrons behave like a wave
- Uncertainty Principle
- Wave-function
- Particle-in-a-Box; quantum numbers
- quantum mechanical tunnelling
- Pauli principle
- the structure of atom and molecules; spectroscopy

Seeing and sensing at the nanoscale:
- Scanning probe microscopy (STM, AFM)
- Mechanically controlled break junctions (MCBJ) and transport regimes
- Electron microscopy (TEM, SEM)
- Fluorescence microscopy (Epi, confocal, TIRF, FRET, DNA PAINT)
- Nanopore technology;

Materials at the nanoscale:
Terminology: dimensionality, allotropy, polymorphisms;

Synthesis and preparation of nanomaterials:
- Top down: milling, exfoliation
- Bottom up: CVD, epitaxial growth, arc discharge, laser ablation
- Nanoparticles: colloidal synthesis, characterization and disaggregation strategies, applications (plasmonics, photonics, catalysis, biolabelling)

Carbon, the amazingly diverse building block:
- The family of carbon nanomaterials (0D, 1D, 2D)
- Structure-property relationship in fullerenes, carbon nanotubes, graphene and nanodiamond
- Characterization techniques
- Applications, e.g. Sensing (mass, pressure, bio), actuation, imaging

Study Goals
By the end of the course, you should be able to:

LO1: Explain the basic concepts and expressions used to describe nanoscale structures and processes;
LO2: Describe the measurements tools developed to probe features and properties of nanoscale materials;
LO3: Discuss the top-down and bottom-up techniques used for the preparation of nanomaterials.

Education Method
Lectures will be given on-site. Mandatory student participation: For each lecture, a team of max. 3 students will be assigned to present a 5 min summary of the previous lecture.
Optional problem sets will be given and discussed on a weekly basis.

Books
The book closest to the lecture is:
Chin Wee Shon, Sow Chorng Haur and Andrew TS Wee,
Science at the Nanoscale - An Introductory Textbook
Pan Stanford Publishing, Singapore, 2010
ISBN: 13 978 981 4241 03 8
ISBN: 10 981 421 03 2
[Concise book with exercises and indications for further readings. Some topics are more detailed and other less than what I can do during the lecture. Some topics are not treated.]

Alternative with more emphasis on Quantum Mechanics:
Introduction to Nanoscience
S.M. Lindsay
Oxford University Press, Oxford New York, 2010
ISBN 9780199544219

The part on Quantum Mechanics can be found in any university level Physics textbook under the rubric 'Introduction into Modern Physics' or similar. Examples are Alonso-Finn Vol. 3 'Quantum and Statistical Physics' or Chapters 34 - 36 of Tipler-Mosca 'Physics for Scientists and Engineers'

E. Meyer, H.J. Hug and R. Bennewitz, Scanning Probe Microscopy - the lab on a tip,
Springer Verlag, Berlin, Heidelberg, New York, 2004.
ISBN: 3 540 43180 2.
[Covers the most important aspects of nanotools, some topics of the course are not treated]

E. L. Wolf, Nanophysics and Nanotechnology,
Wiley-VCH, Weinheim, 2004
ISBN 3 527 40407 4.
[Gives a good introduction to some aspects of nanotechnology, however does not cover the full course and not the full depth.]

Reader
There is no special reader, however, students will be able to download/watch:
- presentation slides;

	- videos shown in the lectures; - supplementary texts on specific topics.
Assessment	The examination will be as follows: Written 3h exam on campus in Q3 (resit in Q4). The written exams counts for 100% of the final grade. Details will be published on Brightspace and discussed in class. The exam will be graded in the ANS plarform and results will be posted on OSIRIS. If less than 10 students register for the exam, an oral exam will be given instead. To enrol for the exam, students must have done a recap in the duration of the course. Enrolment is via OSIRIS. To pass the course, the final grade must be a 5.75 or higher.
Exam Hours	In case of an oral exam, a Doodle poll will be circulated where students will have to register for a specific time-slot (20 minutes) in addition to the regular registration on OSIRIS.
Enrolment / Application	Registration: To participate in this course, registration is via the Brightspace page
Department	ME Department Precision & Microsystems Engineering
Contact	e-mail: S.Caneva@tudelft.nl (preferred) Office: 34.G-1-420 phone: +31 15 27 81673

ME46055	**Engineering Dynamics**	**4**
Responsible Instructor	Dr. F. Alijani	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Multibody Dynamics A(ME41050), Multibody Dynamics B (ME41050), Nonlinear mechanics(ME46000), Nonlinear dynamics (ME46072).	
Course Contents	The dynamic behavior of structures (and systems in general) plays an essential role in engineering mechanics and in particular in the design of controllers. In this master course, we will discuss how the equations describing the dynamical behavior of a structure and of a mechatronical system can be set up. Fundamental concepts in dynamics such as equilibrium, stability, linearization and vibration modes are discussed. If time permits, also an introduction to numerical integration techniques is proposed. The course will discuss the following topics: --Review of the virtual work principle and Lagrange equations --Linearization around an equilibrium position: vibrations --Free vibration modes and modal superposition --Forced harmonic response of nondamped and damped structures -- Introduction to time integration techniques	
Study Goals	The student is able to select different ways of setting up the dynamic equations of mechanical systems, to perform an analysis of the system in terms of linear stability and vibration modes and to properly use mode superposition techniques for computing transient and harmonic responses. More specifically, the student must be able to : 1. Explain the dynamic principle of virtual work and Lagrange equations, and discuss their relation to the basic Newton laws. 2. Describe the concept of kinematic constraints and identify a proper set of degrees of freedom to describe a dynamic system. 3. Use Lagrange equations via employing the minimum set of degrees of freedom to find the governing equations of dynamic systems, and construct these equations for systems with kinematic constraints. 4. Use Hamiltons principle to find the governing equations of motion of dynamics systems. 5. Find equilibrium positions and construct the linearized equations of motion by using different contributions of the kinetic and potential energies. 6. Analyze the linear stability of dynamic systems according to their state space formulation. 7. Explain and use the concept of free vibration modes and frequencies for multi degree of freedom systems. 8. Apply the orthogonality properties of modes to describe the forced response of damped and undamped systems.	
Education Method	Lecture	
Literature and Study Materials	Course material: Lecture notes (available through brightspace) References from literature: 1. M. Géradin and D. Rixen. Mechanical Vibrations. Theory and Application to Structural Dynamics. Wiley & Sons, 2nd edition, 1997. 2. J. Ginsberg. Engineering Dynamics. Cambridge University Press, 2008. 3. D.J. Inman, Engineering Vibration. PrenticeHall, 1996. 4. L. Meirovitch. Principles and Techniques of Vibrations. PrenticeHall, 1997.	
Assessment	written exam +A quiz or an assignment	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	In case an assignment is given- The assignment will involve the use of a Matlab-like software 2. Old course code: WB1418-07	
Department	ME Department Precision & Microsystems Engineering	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realiseable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Contact	s.h.hosseinniakani@tudelft.nl

RO47003	Robot Software Practicals	5
Responsible Instructor	J.F.P. Kooij	
Instructor	G.A. van der Hoorn	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming experience (e.g. python or matlab) at the BSc level, familiarity with the use of functions, variables, and control structures (for-loops, if-statements, etc.).	
Course Contents	This course teaches students the basics for creating and collaborating on academic robotics software. It aims to give students sufficient knowledge and experience such that they can proceed learning more about robotics programming by themselves as needed. The course does this by providing an overview of key technologies, and letting students develop basic skills often used in real world robotic software development. The main objective is to obtain hands-on experience - beginners level - with Linux, Git & Gitlab, C++, ROS, and integrate with common external libraries such as OpenCV and PCL. This course does not teach new robotics theory (e.g. perception, planning, control, etc.) which is covered in other courses. The allotted 5 ECTS means that students are expected to invest approximately 11 hours/week outside of contact hours. Be aware that active participation in all the lab and peer-review assignments is required, hence it is not recommended to do this course if you cannot commit to a regular time investment over the whole education period.	
Course Contents Continuation	Linux: Although Windows is the most prevalent operating system on desktop PCs, Linux became very popular for embedded systems such as robots. Developing for embedded Linux systems is most easily done in Linux itself, and this course aims to familiarize students with the use of Linux on the desktop and terminal. It includes the following topics: OS architecture, File system, Shell, Scripting. Git/gitlab: All exercises will be submitted using Git, a widely used version control system to manage and collaborate on coding projects. During the course, students will practice basic tasks with git, such as solving code conflicts, and merging the work of lab partners. Using an issue tracker and giving constructive feedback to peers is also an important aspect that will be practiced. C++: C++ is one of the most relevant programming languages for robotics. Being an object oriented language brings great advantages compared to its predecessor C. This practical gives students practical knowledge on C++ programming. The course encompasses the following topics: Introduction to C++, common compilation tools, basic programming in C++, classes and objects. ROS: The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries, and conventions that aim to simplify the task of creating complex and robust robot behaviour across a wide variety of robotic platforms. During the course, students will learn to use existing ROS tools, create their own software components in C++, integrate them into an application, and test their solutions using a physics simulator. others: You will also get some practice using external libraries in your robotics project, such as OpenCV or PCL. OpenCV is a computer vision library designed for computational efficiency and with a strong focus on real-time applications. The Point Cloud Library (PCL) is an open-source project for point cloud processing. Note that this course is *not* about understanding the methods or theory on which these libraries are based, but rather on integrating the provided functionality in novel software components.	
Study Goals	--------------------------------------- Knowledge, insight, judgment and skills --------------------------------------- After completing this course, students will be able to 1. use the command line on a Linux system, perform basic file operations and job management tasks, and automate tasks with shell scripts; 2. use Git to commit and integrate incremental code changes together with a lab partner, and resolve any code integration conflicts; 3. use a Gitlab server to exchange code changes, and use its issue tracker to provide and receive feedback on each other's code; 4. solve programming tasks in C++, compile the code using Make, debug compile-time, link-time and run-time problems; 5. create new C++ ROS packages, use common ROS development tools, and utilize the ROS API in their C++ programs; 6. investigate and integrate external libraries into their own ROS components, such as OpenCV or PCL; 7. add code comments and basic documentation to their code, find and interpret documentation of external libraries; ------------------------------------- Transferable and interpersonal skills ------------------------------------- 8. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback; 9. collaborate and coordinate with lab partner on coding tasks.	
Education Method	Lectures, which will introduce new course content (2 hours per week). Interactive Sessions will provide class wide feedback on past lab assignments, tutorials, quiz exercises, Q&A (2 hours per week) Practicum session, during which there will be TAs around to help you understand the lab assignments (2 hours per week) Robotics Software Practicals focuses on improving programming and collaborative software development skills. The main method to develop these skills is by gaining hands-on experience through lab assignments, and receiving feedback on the submitted work. Lectures will introduce the various topics, provide context and tips, and will discuss common pitfalls and solutions. Still, the nature of the assignments means that students are expected to invest sufficient time for self-study and practice outside the fixed lab and lecture contact hours. Lab assignment There are 4 obligatory lab assignments, each covering two weeks, namely 1. Linux, Git (Formative) 2. C++ programming (Formative) 3. ROS (Formative) 4. Final Assignment (Summative, demonstrates capacity to use Linux, Git, C++, and ROS). A manual will be provided for each lab assignment before the start of the first practicum session for the assignment. The manual will contain detailed instructions, guidelines, optional exercises, and an assignment that must be submitted to an online repository for your lab group.	
Literature and Study Materials	1. A Gentle Introduction to ROS, Jason M. OKane. http://www.cse.sc.edu/~jokane/agitr/ 2. Provided lab manuals, and lecture slides that will be made available on Brightspace	

Practical Guide	For the practicum assignments, bring a reasonably recent laptop with a x86-64 CPU (i.e. a normal" Intel or AMD CPU; but NOT the new MacBook Air with M2 chip, NOT a Chromebook, etc. These use different CPU architecture which means that the software we provide will not run! A MacBook with an Intel x86 CPU is fine). A laptop with an Nvidia GPU would make the robot simulations run even smoother. Please note the following lab rules: The first three lab assignments are made in groups of two, as you will learn to collaborate on code with your lab partner using gitlab. You can find a lab partner using the special Brightspace forum, but of course also by talking to others after the first lecture, or by looking for a Brightspace lab group with a single enrolled person. To enrol in the practicum, and therefore participate in the course, you and your partner *must* enrol on Brightspace in a lab group before the lab enrol deadline (this deadline will be announced in first lecture). If you plan to drop the course, inform the lab coordinator in time before the start of the next lab assignment, but while you are still enrolled, you should finish the ongoing lab assignment with your lab partner. The scheduled practicum sessions are intended for you to obtain help on the assignments from the TAs. Outside the practicum sessions, you can post questions on the TA forum on Brightspace. To submit your work, you and your partner must push your contributions to a provided gitlab repository. Note that using git and the gitlab webserver for code collaboration is part of the learning objectives of this course. You are responsible for testing the code that you hand-in via gitlab, so make sure that you have tested your work thoroughly, and that the correct version has been uploaded to your gitlab repository. For the first three lab assignments, there will be a peer-review task after the lab deadline has passed. You will learn during the course how to provide feedback, and how to use the gitlab issue tracker to manage issues and comments. The lectures will afterwards discuss positively peer-reviewed lab solutions with all students for additional feedback by a teacher. Active participation in each lab assignment is required to participate in the next assignment. Active participation entails that you submit your solution before the given deadline, and that you complete the required peer-review tasks. You may only use the gitlab code repository that the lab coordinator will create for your group. You are *NOT* allowed to put your coding homework on github.com, gitlab.com, bitbucket.com, google drive, dropbox, etc. (neither with public nor private settings!) TU Delft ethical guidelines apply to reports, code, etc. If your name is on it, you claim it is your work, do *not* copy parts of other student groups code. You are *not* allowed to share your code, solutions with others apart from your lab partner and TAs. Generally, do *not* show your groups work to other students. Plagiarism will be reported to exam committee.
Assessment	# Assessment overview Formative: First three lab assignments, peer-review tasks, discussion on common mistakes and solutions of past hand-in assignments during the lecture by teacher, multiple-choice practice questions. Summative: The final lab assignment, and digital exam. # Grade calculation You final course grade (minimum is a 6 to pass) determined by Final lab assignment (grade, weight 25%, minimum a 5.5 to pass), Digital exam (grade, weight 75%, minimum a 5.5 to pass) Any penalties (see Deadlines below) A resit will be available for the exam, but not the final lab assignment. # Lab assignments You must enroll for the obligatory lab assignments before the announced enrollment deadline (given during first lecture), so that the lab coordinator can fix the groups and setup everybodys code repositories in time. If you do not enroll, you cannot make the lab assignments, or make the exam. In the first three assignments you must collaborate with a lab partner. Your group will receive and give formative feedback through student peer-review (what goes well, what to improve). The fourth and final lab assignment is made individually, and you receive a grade (weights 25% for final grade), based on a rubric for the quality of the solution, code, documentation, and use of Git. This assignments requires you to apply all knowledge from the previous lab assignments to solve a well-defined robotics programming task. Your (groups) lab work is evaluated on the following three Lab Criteria (LC): - LC1: solution quality (does the solution solve the given task?) - LC2: code quality (are there any bugs or problems with the code?), and - LC3: documentation quality (is there a clear readme, does the code contain sensible comments?). # Deadlines, submission requirements, and penalties While most of the lab assignments are formative, your active participation is required and there are strict submission and peer-review requirements: - LC4: Each partner must fulfil each labs minimum submission requirements regarding use of Git, and Gitlab, file layout, and build instructions, before the lab deadline. Each assignments unique submission requirements will be listed in the corresponding lab manual. - LC5: You must participate in the peer-review tasks for the first three lab assignments, and complete these before the given peer-review deadlines. Failing to comply with these criteria will result in penalties on your final *course* grade: - After lab submission deadline: -1 grade point per day for not completing the assignments minimum submission requirements, with a maximum of -2 per assignment. Not enrolling in time before the lab automatically results in -2. - After peer review deadline: immediate -0.5 grade point for not fully completing the peer-review task (only applies to those enrolled for the lab). NB: - These criteria apply to both lab partners individually. If you comply with the criteria but your partner does not, only your partner will receive the penalty. - Penalties apply to your final course grade, so if you do not submit for one lab (-2 penalty) you could only complete the course with an 8 at best. - There are NO redos or extra assignment opportunities to remove a given penalty. You will have to redo the lab next year and comply with the deadlines then for a clean slate. # Exam There will be an individually-made digital exam, for which you will receive a grade (weights 75% for final grade, a minimum of 5.5 to pass). Questions are based on content discussed in the lectures, and on practical situations that you have encountered during the lab assignments. The exam will also require you to reflect on and refer to your own coding solutions for the final lab assignment, therefore to participate in the exam you *must* have participated in the final lab assignment. Practice questions will be provided before the exam so you can assess your own understanding first.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on

	https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47004	Machine Perception	5

Responsible Instructor	Prof.dr. D. Gavrila
Practical Coordinator	R.M. Ensing
Contact Hours / Week x/x/x/x	0.6.0.0
Education Period	2
Start Education	2
Exam Period	2 3
Course Language	English
Course Contents	This course provides an overview of machine perception techniques in robotics, with a focus on intelligent vehicles. Interested students outside ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties) with the proper background (see Prerequisites below) are encouraged to attend. Introduction Machine perception in robotics Course organization Sensor Overview (camera, radar, LiDAR, tactile) 3D Machine Vision Perspective camera model Extrinsic and intrinsic camera transformations Stereo vision Radar Sensing and Processing Principles of FMCW Radar Distance, velocity and angle measurements Use in object detection Lidar Sensing and Processing Point cloud representations Use in object detection Tactile Sensing and Processing Piezoresistive, capacitive, piezoelectric and optical sensing Contact and slippage Shape extraction and elasticity Visual Object Detection and Classification Detection vs. classification Object proposals Handcrafted features (e.g. HOG) & classification (e.g. linear SVM) End-to-end learning: neural networks, deep learning Performance metrics: confusion matrices, precision vs. recall, ROC curves State Estimation Bayesian Filtering Kalman Filtering Particle Filtering Object Tracking Data Association Track Management Self-Localization & Sensor Fusion Absolute vs. relative localization Ego-motion compensation (e.g. odometry, ICP algorithm) Extrinsic sensor calibration Environment representations (grids, voxels) The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).
Study Goals	After completing this course, students will be able to: explain the role of Machine Perception (MP) in Robotics, and describe possible applications explain the measurement principles of the relevant sensors explain the principles of well-established methods for low- to high-level sensor processing analyze a MP problem, consider available sensor and computational resources, and select the appropriate MP methods to apply write Python code in relevant frameworks to visualize data and implement MP methods perform MP experiments, evaluate the results, and draw sound conclusions
Education Method	Lectures (2-4 hours per week) Practicum (2 hours per week) Formative feedback: Practica - Pairwise collaboration (alternating random groups) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle run the software locally on their own computers (laptop/PC). However, we provide no support for setting up the necessary software environment (e.g. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.
Course Relations	This course, RO47004 Machine Perception *cannot* be selected in combination with ME41106 Intelligent Vehicles, due to the content overlap between the courses. This course, RO47004 Machine Perception provides the basics for the follow-up course "RO47020 Advanced Machine Perception".
Literature and Study Materials	Slides Hand-outs Jupyter notebooks
Prerequisites	Basic linear algebra and probability theory Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope. Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent) It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this

	catching up will likely result in an increased study load (e.g. 1-2 EC).
Assessment	Final practicum assignment (50%) - Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	For more information about this course, see the slides of the Introduction lecture from past academic year at http://intelligent-vehicles.org/teaching/courses/ . TU Delft students might also be able to view the lectures from past academic year on Collegerama.
Department	ME Department Cognitive Robotics

RO47005	Planning & Decision Making	5
Responsible Instructor	Dr. J. Alonso-Mora	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an overview of motion planning and decision-making techniques and their practical application in robotics. Planning and Decision-making are critical components of autonomy in robotic systems. These components are responsible for making decisions that range from path planning and motion planning to coverage and task planning to taking actions that help robots understand the world around them better. This course studies underlying algorithmic techniques used for planning and decision-making in robotics and examines case studies in ground and aerial robots, mobile manipulation platforms and multi-robot systems. The students will learn the algorithms and implement them in a series of programming-based projects. In particular, we will first cover the fundamentals of planning (workspace, configuration space, representation of obstacles and robot models). We will then cover a broad set of methods, which include reactive methods and feedback control; discrete, combinatorial and probabilistic planning; planning under differential constraints; planning under uncertainty; learning in planning; and planning for multi-robot systems. Finally, we will briefly touch upon task assignment and vehicle routing. Course structure: Introduction Fundamentals: 3D rotations, configuration space, algorithm properties Differential constraints: recap of modeling & control of manipulators, ground robots, aerial robots Graph-search fundamentals (depth/breadth first, Dijkstra, A*, heuristics) Combinatorial algorithms (cell decomposition, shortest-path roadmaps, motion primitives) Sampling-based algorithms (PRM, RRT, RRT*) Reactive algorithms: collision avoidance, potential fields, velocity obstacles Kinodynamic planning Trajectory optimization (fundamentals, global, local, Model Predictive Control) Planning under uncertainty (Markov Decision Processes, learning) Multi-robot motion planning (joint configuration, optimization-based, coverage) Task assignment (Hungarian, auction, linear program) Intro to vehicle routing	
Study Goals	After completing this course, students will be able to: explain the role of Motion Planning and Decision-Making (PDM) in Robotics, and describe possible applications identify different classes of robotic systems, the associated mathematical models for their kinematics, dynamics and motion and the relevant spaces for PDM, such as the configuration and state spaces describe and compare algorithms for motion planning and decision-making of mobile robots and multi-robot systems, including: reactive algorithms, optimization-based algorithms, combinatorial algorithms, sampling-based algorithms and learning-based methods analyze a PDM problem, consider constraints, objectives and select the appropriate PDM methods to apply design and implement motion planners of moderate complexity to solve motion-planning tasks and navigate mobile robots perform PDM experiments, evaluate the results, and draw sound conclusions present the findings of a PDM experiment clearly in a report using graphs and scientific English understand PDM experiments descriptions found in scientific literature, and reproduce results	
Education Method	Lecture hours/week: 4 Q&A sessions/week: 2 Practical assignments and project (home) hours/week: 6 Self-study hours/week: 4	
Books	Planning Algorithms, S. LaValle http://planning.cs.uiuc.edu/ And additional papers/references provided by the lecturer.	
Prerequisites	Linear Algebra Programming in Python Rigid body dynamics (recommended) Dynamics & Control (recommended) Probability Theory (recommended)	
Assessment	Practicum assignments (10%) Project (40%) Collaboration in groups of 3 students - No resit (repeat the following year) Written Exam (50%) individual Knock-out criteria - Minimum grade for the project is a 5.0 - Minimum grade for the exam is a 5.0	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Cognitive Robotics	

RO47017	Vehicle Dynamics and Control	5

Responsible Instructor: B. Shyrokau

Contact Hours / Week x/x/x/x: 0.0.0.4

Education Period: 4

Start Education: 4

Exam Period: 4
5

Course Language: English

Course Contents

This course provides fundamental knowledge on vehicle motion, vehicle-road interaction and the main technical principles and aspects of automotive control systems. This course discusses vehicle modelling, controller design and testing procedures to evaluate vehicle motion and automotive control systems.

Tire dynamics and modelling
Wheel slip, slip angle, tire force / moments, influence of camber angle, pressure, temperature effect, combined slip, friction circle, transient behavior
Tire modelling and wheel dynamics

Lateral motion and vehicle handling
Kinematics, understeer gradient, bicycle model, steady-state handling
Transient motion, frequency response, handling testing, vehicle stability

Antilock braking system (ABS) and vehicle dynamics control (VDC)
General concepts of ABS and different control principals: rule-based, mixed slip-deceleration, advanced control techniques
General concept of VDC, yaw rate based controller, body slip angle based controller, integrated vehicle control using control allocation

Planar motion
Planar vehicle modelling
Static equilibrium, lateral load transfer, roll center and axis, roll stiffness, anti-roll bars
Longitudinal load transfer, pitch center, anti-dive, anti-lift

Steering & assist logic
Wheel kinematics, steering system modelling, column- and rack-assist
Steering assist logic, steering feel and assessment

Path-following and evasive control of automated vehicles
Vehicle road model, error definition, an overview of path-following controllers, PD controller
Evasive maneuver, advanced control techniques, MPC-based evasive steering assist

Vertical motion and comfort
Introduction to vibration, quarter car, 2DoF and 3DoF models, Ride comfort
Suspension control: skyhook, groundhook, and mixed control

Vehicle state estimation
General concept of VSE, vehicle velocity estimation, Kalman filtering (linear and extended), wheel load reconstruction methods

Final presentations of the group project

Study Goals

After successfully completing this course, you will be able to:
explain the tire-road interaction and be able to implement the tire model and compare it with the established one
describe the concept of vehicle handling and lateral stability and derive the equations of motion of a nonlinear vehicle model
interpret the load transfer effects and implement four-wheel planar vehicle model including the wheel kinematics, steering system and steering assist logic
explain the concepts of wheel slip control, vehicle stability control, automated lateral control and vehicle state estimation
implement control strategies for vehicle stability control (PID with gain scheduling or LQR) and automated lateral control (MPC)
derive the equations of vertical motion of a quarter-car vehicle model and explain the influence of vehicle and road parameters on vehicle vibration behaviour

Education Method

Lectures (4 hours per week)
Office hours (2 hours per week)
Group project & presentation

Prerequisites

Basic programming experience (e.g. Matlab / Simulink) at the BSc level
Dynamics and control at the BSc level
knowledge about Model Predictive Control is recommended

Assessment

Homework assignments (25%) individual
Group project (25%) collaboration in groups of 4-5 students
Written exam (50%) individual

The examination is a closed-book digital exam on campus including multi-select and open questions.

The grade for the homework assignments will be the average grade of the 3 assignments.

Knock-out criteria
o Minimum average grade for homework assignments is a 5.0
o Minimum average grade for group project is a 5.0
o Minimum grade for the exam is a 5.0
o Final grade minimum is a 5.75

Enrolment / Application

Registration:
To participate in this course, registration via MyTUDelft is not needed.
More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses.

To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.

*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the

	OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics
Contact	Barys Shyrokau, b.shyrokau@tudelft.nl

RO47019	**Intelligent Control Systems**	**4**
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Lineair Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	Dr. O.A. Soloviev	
Responsible Instructor	Dr.ing. C.S. Smith	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42061	Nonlinear Control Systems	3
Responsible Instructor	Dr.ir. M. Jafarian	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Linear state space models (e.g. SC42015 Control Theory) - Basic knowledge of (multivariable) calculus, differential equations and linear algebra	
Course Contents	- Fundamentals of nonlinear systems (basic properties, solvability, types of equilibrium points) - Analysis of planar systems (graphical representations, periodic orbits, elementary bifurcations) - Stability theory for autonomous nonlinear systems (Lyapunov's methods, invariance principle) - Advanced stability theory (time-varying systems, input-to-state stability, passivity) - Observers (local, global and high-gain, extended Kalman filter)	
Study Goals	Students can: - determine if a system is nonlinear, time-invariant, autonomous, etc. - determine time intervals over which a system is solvable - sketch and interpret graphical representations of 2d systems - check for the existence or absence of periodic orbits in 2d systems, compute bifurcation points, and classify elementary bifurcations - determine if equilibrium points of autonomous nonlinear systems are (asymptotically, exponentially, ...) stable - determine if equilibrium points of certain classes of non-autonomous systems are (asymptotically, exponentially, ...) stable - construct observers for certain classes of nonlinear systems	
Education Method	Lectures, problem sets and instruction sessions.	
Literature and Study Materials	- H. K. Khalil: "Nonlinear Control", Pearson, 1st Global Edition, 2015. ISBN 9781292060507. - Additional handouts	
Assessment	Written exam, depending on the COVID-19 situation either on campus or online.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ing. C.S. Smith	
Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC4115	
Department	ME Department Delft Center for Systems and Control	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). This course provides an introduction to modeling, analysis, and control of hybrid systems. The main topics considered are: * General introduction, examples of hybrid systems and motivation * Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) * Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) * Control of hybrid systems (switching controllers, model predictive control) * Verification and tools	
Study Goals	After successfully completing the course, you will be familiar with the main aspects and characteristics of hybrid systems, you have insight in trade-off modeling power versus decision power in the context of analysis and control of hybrid systems, and you will have obtained an overview of modeling, analysis, and control methods of tractable classes of hybrid systems. In addition, you will be able to apply hybrid systems modeling and control to simple case studies.	
Education Method	lectures + mandatory assignment Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.	
Assessment	Written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: partial marks for exam or assignment do not carry over from one academic year to the next.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four written assignment reports, each representing 25% of the final grade. 2. In order to complete the course, each assignment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42110	Dynamic Programming and Stochastic Control	5
Responsible Instructor	M.A. Sharifi Kolarijani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Solid knowledge of undergraduate probability (e.g., conditional distribution, expectation, variance), linear algebra (e.g., eigendecomposition and Jordan forms), and calculus (e.g., gradient and extrema of a function). Mathematical maturity and the ability to provide precise and rigorous arguments are also important. A class in analysis will be helpful, although this prerequisite will not be strictly enforced.	
Course Contents	The course covers the basic models and solution techniques for stochastic optimal control problems and sequential decision-making under uncertainty. The course starts with a class of stochastic processes known as Markov chains. We then extend the model to include control variables (Markov Decision Processes, or in short MDPs) and study optimal control of a dynamical system over a finite number of stages. We also discuss an important class of problems known as optimal stopping problems. Different applications from engineering and finance to operations research and management science will be covered.	
Study Goals	After successfully completing this course, you will be able to - model real-world control and decision-making problems under uncertainty and in dynamic environments using the framework of discrete-time Markov decision process; - compute relevant technical concepts including hitting probabilities, expected hitting times, invariant distributions, and the long-run proportion of time spent in a given state; - formalize the optimal decision-making problem by deriving the respective dynamic programming principle; - use the dynamic programming algorithm to find useful properties of the desired solution and solve the optimal decision-making problem.	
Education Method	Lectures and tutorials	
Literature and Study Materials	- The lecture notes of the course (available on Brightspace) - Dynamic Programming and Optimal Control, 3rd edition, D. Bertsekas, Athena Scientific, 2005 - Introduction to Probability, D. Bertsekas and J. Tsitsiklis, Athena Scientific, 2002	
Assessment	Written exam (If on-campus exams will not be possible, then all exams will be organized online via Ans)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	The students can reach the course staff via email regarding administrative issues.	

SC42115	Internship	6
Responsible Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The internship is optional (a free technical elective) performed at a company or research institute (not a university) and has to be chosen with the approval of the responsible teacher.	
Study Goals	The students have demonstrated their capability, independently and in consultation with specialists, to define, limit, solve and discuss systems and control problems as defined in the internship project description. The students have proven to be capable of communicating about their Internship research project both through an oral presentation and a report. The students have demonstrated their capability to consider and discuss the technological, ethical and societal impact of their internship work. The students have shown their life-long learning competence by investigating the scientific publications related to the problems investigated in their internship thesis and processing this information in their thesis.	
Education Method	Project	
Assessment	Report	
Enrolment / Application	* Students find an interesting internship proposal. * For internships in the Netherlands, the form Internship agreement UNL, including Annex 1, needs to be filled out together with the company / research institute. For internships outside of the Netherlands, the form Internship application form needs to be filled out together with the company / research institute. These forms contain information about problem statement, proposed approach and expected outcome. * Responsible teacher checks the intake form and, if OK, approves the internship. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Mathematics Matlab Optimalisation	
Department	ME Department Delft Center for Systems and Control	

SC42130	Fault Diagnosis and Fault Tolerant Control	4
Responsible Instructor	Dr. R. Ferrari	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Control Systems Design, System Identification, Signal Processing, Statistics and Probability	
Summary	In critical systems such as transportation systems, power plants, distribution networks or life-support systems it is not enough to design a control system that meets the required performance levels. It is also of paramount importance to guarantee that such requirements, or a subset of them, are met also during non ideal conditions such as in the presence of physical failures or cyber attacks. Most of all, safety must be guaranteed at all times. This can be obtained by designing into the control system additional functionalities that will allow to detect the occurrence of faults or cyber attacks, and deploy adequate countermeasures, which may include the reconfiguration of the control system.	
Course Contents	Faults and failures in dynamical systems; Cybersecurity and cyber attacks to Industrial Control Systems; Components and Services model; Availability and Reliability; FTA (Fault tree Analysis); FMEA (Failure Modes and Effects Analysis); Signal based methods for Diagnosis; Model based methods for Diagnosis; Residuals computation and evaluation; Learning of fault model; Fault Tolerant Control (Model Matching, Virtual Sensor and Actuator, Adaptive Control)	
Study Goals	after the course the student should be able to: 1) analyse the structure of a system and identify main components, the services provided by each and carry out a Fault Modes and Effects Analysis; 2) analyse and model faults or cyber attacks that can occur in a given dynamical system; 3) design an algorithm for detecting, isolate and identify them; 4) design a policy for reconfiguring the control system in order to accommodate them. 5) analyze the vulnerability of an industrial control system to cyber attacks and design prevention and detection algorithms	
Education Method	in person lectures, in-class exercises, homework	
Literature and Study Materials	Recorded lectures, Slides, Example exercises with partial solutions, Suggested textbook: Blanke, M., Kinnaert, M., Lunze, J., Staroswiecki, M., & Schröder, J. (2006). Diagnosis and fault-tolerant control (Vol. 691). Berlin: Springer.	
Assessment	Assignments 30%, final exam (written) 70%. The grade for the assignments will be the average grade of all the assignments. To pass the course, the final grade must be a 5.75 or higher. The grade for the exam is not allowed to be lower than a 5.0.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Design Group work Mathematics Matlab Mechatronics Modelling Numeric Methods Process Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42140	Signal Analysis & Learning for Two-Dimensional Systems	3
Responsible Instructor	Dr.ing. R. van de Plas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WB2235 (Signaalanalyse) or equivalent.	
Course Contents	Main concepts from classical signal and systems theory are extended from the 1-D time-centric case to the N-D multidimensional case, with specific focus on 2-D spatial signals and systems. The generalization of sampling theory to two-dimensional systems provides the link between continuous-space signals and their discrete-space measurements, and includes multidimensional views on the Shannon-Niquist theorem, interpolation, and aliasing. The translation of imaging data into alternative transform domains by means of the wavelet, cosine, or Fourier transforms, and the usage of 2-D filters will be introduced as means for image enhancement, de-noising, compression, and the extraction of relevant features. Once filtering and transformation have delivered a more advantageous representation of the imaging data, the course follows up with elementary machine learning methods to mine these often multivariate and multidimensional signals for specific applications. This includes supervised learning methods aimed at answering specific hypotheses or questions, and performing specific image recognition tasks. Within this imaging context, the classical modeling trade-off between overfitting and generalization is addressed, and statistical resampling methods are introduced as ways to manage model selection. The course furthermore includes unsupervised learning methods aimed at open-ended exploration of imaging data, to reveal underlying structure and trends. The main topics of the course are: 1. Two-Dimensional Signals and Systems; 2. Two-dimensional Sampling Theory; 3. Two-Dimensional Discrete-Space Transforms; 4. Signal Processing for Imaging Systems; 5. Supervised Learning (regression, classification); 6. Assessment and selection of learned models (bias-variance trade-off, cross-validation, bootstrap); 7. Unsupervised Learning (clustering, principal component analysis)	
Study Goals	See Course Contents topics.	
Education Method	Lectures and student presentations.	
Literature and Study Materials	- Woods J.W., Multidimensional Signal, Image, and Video Processing and Coding, 2nd Ed., Academic press (2011). - Hastie T., Tibshirani R., and Friedman J., The Elements of Statistical Learning - Data Mining, Inference, and Prediction, Second Edition, Springer Series in Statistics (2009).	
Assessment	Final grade consists of the students presentation grade (40%) and exam grade (60%). The final exam is an oral exam if practically feasible, or a written exam otherwise.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42160	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	Dr. N.J. Myers	
Instructor	G. Joseph	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the L0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The L0 norm is shown to be NP-hard, and several classical approximate algorithms like L1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Discuss the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Discuss the concept of sparsity and compressive sensing from an optimization perspective. 4. Analyse different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP; EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.;	
Books	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (25%), a mini project (15%), and homework submission (10%). The course is closed by a project report, presentation, and Q&A (50%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42170	Probabilistic Sensor Fusion	3
Responsible Instructor	Dr. R.T. Rajan	
Responsible Instructor	Dr. M. Kok	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	ET4386: Estimation and Detection Theory OR SC42025: Filtering and Identification	
Course Contents	The objective of this course is to apply probabilistic sensor fusion methods to real data. The main focus is both on implementing these methods for real-life data sequences and on understanding, evaluating and interpreting the estimation results and the workings of the algorithms. The course focuses on nonlinear filtering techniques (extended and unscented Kalman filtering, and particle filtering), Gaussian processes, as well as distributed versions of these algorithms.	
Study Goals	At the end of the course you should be able to: - Apply EKF / UKF / PF / Gaussian processes to a real-life data sequence - Assess the results and compare between different algorithms and different settings - Discuss and compare the results	
Education Method	Lectures and practical assignments	
Literature and Study Materials	Lectures, accompanying lecture slides and material accompanying the practical assignments.	
Assessment	The final assessment is based on practical hand-in assignments and a final discussion and presentation of the results.	
Remarks	The practical assignments will be implemented using Python.	
Department	ME Department Delft Center for Systems and Control	

SC47500	Tensor Networks for Green AI and Signal Processing	4
Responsible Instructor	Dr.ir. K. Batselier	
Instructor	Dr. B. Hunyadi	
Contact Hours / Week x/x/x/x	HC 2.0.0.0 PCP 4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course provides a solid mathematical foundation to tensor networks and discusses their use in machine learning and signal processing. Tensors are multi-dimensional generalization of matrices. Real-life data often comes in high dimensions, such as of a video stream, or multichannel EEG recorded from multiple patients under various conditions. Tensor-based tools are crucial to efficiently represent and process this data. Alternative (matrix-based) solutions artificially segment such high-dimensional data into shorter one- or two-dimensional arrays, causing information loss by destroying correlations in these data. The key idea of tensor networks is to write a tensor in terms of much smaller tensors that represent these correlations among the different dimensions. On one hand, such representations can reveal hidden structure in the data and can even lead to decompositions with physically interpretable terms. Moreover, in this way, the original tensor never needs to be kept explicitly in memory. This latter aspect becomes especially interesting considering the ever-increasing dataset sizes brought on by recent advances in sensor and imaging technology. Computations based on tensor networks enable faster training and inference of AI models based on such large datasets. Therefore, tensor networks facilitate the shift towards the direction of Green AI. Green AI aims to decrease the environmental footprint of AI computation and increase the inclusivity of AI research by becoming more independent of expensive computer hardware needed for training. The first introductory lecture gives a motivation to the use of tensor networks. Various compelling examples from the teachers own research experience will be given. Next, the basic notation and the use of tensor diagrams will be presented, and important matrix and tensor operations will be explained, such as various products (inner, outer, Kronecker, Khatri-Rao, etc.), norms, reshaping (matricization, vectorization). These operations and concepts are crucial to understand how matrix decompositions generalize to tensor methods. This leads to the core of the course content. The most well-known tensor network is the singular value decomposition (SVD) of a matrix. The following two lectures present how the SVD can be generalized to tensors in two different ways, one leading to the multi-linear singular value decomposition (MLSVD), and the other leading to the canonical polyadic decomposition (CPD). The properties, computation and applications of both generalizations will be discussed. The third tensor network structure that generalizes the matrix SVD are tensor trains (TT). Their properties, computation and applications will also be discussed. TTs are particularly well-suited to represent large-scale vectors and matrices as most of the common linear algebra operations can be done with TTs. Finally, the last two lectures focus on the application of tensors to various signal processing (e.g. blind source separation, data fusion) and machine learning problems (e.g. kernel methods). For this, we will introduce advanced concepts such as different tensorization techniques, constrained and coupled decompositions. Throughout the course, various applications will be presented. From the field of biomedical signal processing this includes epileptic seizure localization in EEG, fusion of simultaneous EEG-fMRI, blind deconvolution of brain responses in functional ultrasound during visual information processing or prostate cancer detection using ultrasound. The course explains how tensor networks pave the way towards Green AI. This is done by showing how machine learning methods like kernel machines, artificial neural networks, and Gaussian processes can be made more efficient and tractable for big data and higher dimensional problems.	
Course Contents Continuation	The course consists of 8 2-hour lectures of theory, including examples and applications. There will be 4 2-hour computer lab sessions that discuss practical implementation and applications. These lab sessions help student complete their computer assignments, which are an important part of their assessment. The examples in the assignments are taken e.g. from biomedical applications (ECG, EEG data), computer vision, etc.	
Study Goals	After the course, the student is able to: Explain different tensor networks (TN) Implement basic tensor networks Formulate a basic data science / signal processing problem as a TN Make informed choices in the practical use of TNs, such as rank or initialization method. Formulate research questions for their own (master thesis level) research or independent literature study in the field and its applications, i.e. tensor-based (biomedical) signal processing, image analysis, control, etc.	
Education Method	8 lectures and 4 computer sessions	
Literature and Study Materials	The course covers chapters Part I and selected chapters of Part II of the book below. Further applications and examples will be provided from recent scientific literature and the teachers own research experience. Andrzej Cichocki, Namgil Lee, Ivan Oseledets, Anh-Huy Phan, Qibin Zhao and Danilo P. Mandic (2016), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 1 Low-Rank Tensor Decompositions", Foundations and Trends® in Machine Learning: Vol. 9: No. 4-5, pp 249-429. http://dx.doi.org/10.1561/2200000059 Andrzej Cichocki, Anh-Huy Phan, Qibin Zhao, Namgil Lee, Ivan Oseledets, Masashi Sugiyama and Danilo P. Mandic (2017), "Tensor Networks for Dimensionality Reduction and Large-scale Optimization: Part 2 Applications and Future Perspectives", Foundations and Trends® in Machine Learning: Vol. 9: No. 6, pp 431-673. http://dx.doi.org/10.1561/2200000067	
Assessment	The students will complete 4 assignments. Each assignment will be discussed during the corresponding computer lab session. The first 3 assignments are common for all students, and they are completed individually. The final assignment is executed in pairs, and the topic is chosen from a list of several options, depending on the interest of the students. The students will submit their code as well as a report about the assignments. The scores of the assignments contribute to final grade with the following weights: 5% / 10% / 10% / 75% and students need to pass each individual assignment. The final assignment will also be defended with an individual oral examination, where theoretical questions related to the project will be asked. In case of a retake a new final assignment has to be completed.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is (G1+G2+2*G3)/4, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	mr.ir.drs. V.N.S.R. Dwarka	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Dr.ir. S. Jain	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High -resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	The final grade of the course consists of the following components: - Homework assignments - Group assignment - oral exam Final grade calculation: Provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: (G_1 + G_2 + 2*G_3)/4. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments: resubmit deliverable - Group assignment: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. C.A. Urzúa Torres	

WI4221	Control of Discrete-Time Stochastic Systems	6
Responsible Instructor	Prof.dr.ir. J.H. van Schuppen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	- Required: Mathematical analysis as provided at the batchelor level; and linear algebra of the batchelor level; - Strongly advised: (1) knowledge of elementary probabily theory; (2) control and system theory of deterministic linear systems.	
Course Contents	Discrete-time stochastic systems, distributions and invariant measures. Stochastic realization. Control with complete observations, optimal control theory, dynamic programming for finite and infinite horizons. Kalman filtering and special cases of filtering of stochastic systems. Control with partial observations, separation property. Control of networked stochastic systems and decentralized control.	
Study Goals	By the end of the course, a course participant should be able: LO1. To analyse the conditional expectation of a Gaussian random variable conditioned on another such variable, of a finite-valued random variable conditioned on another such variable; and the concept of conditional independence. LO2. To analyse elementary properties of stochastic processes. LO3. To evaluate properties of a Gaussian stochastic system, of a finite stochastic system, and of a general stochastic system, in particular the forward and the backward system representations of such systems and their relations; and the concepts of stochastic observability and of stochastic co-observability. LO4. To analyse the results of a stochastic realization of a stationary Gaussian process, in particular the equivalence conditions of minimality of such a realization and the formulation of the set of equivalent minimal realizations; the results of a realization of a deterministic linear system, and the concept of dissipativity of such a system. LO5. To analyse a stochastic control system, in particular the stochastic controllability of such a system. LO6. To evaluate the dynamic programming procedure, the value function, and the optimal control law, for optimal control of a completely-observable stochastic system, both on a finite horizon and on an infinite horizon. LO7. To evaluate the Kalman filter of a Gaussian stochastic system, the Kalman predictor of a Gaussian stochastic system, and the filter of a finite stochastic system. LO8. To evaluate the dynamic programming procedure, the value function, and the optimal control law, for control of a partially-observable stochastic system both on a finite horizon and on an infinite horizon. LO9. To understand several of the research issues of control of a distributed or a decentralized stochastic control system.	
Education Method	Lectures/exercises	
Literature and Study Materials	Notes for this course can be obtained from the instructor.	
Assessment	The final grade of the course consists of the following components: - Weekly homework sets (50%); - An oral exam to be carried out after the last lecture of the course (50%). Final grade calculation: (0.5 * the average grade of the weekly homework sets) + (0.5 * the grade of the oral exam) The homework sets count for 50% if solutions of all sets were sent to the lecturer (proportionally if fewer solutions were provided). Repair possibilities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5,	

	sub 5., for: - Homework sets: submit 1 extra homework set. - Oral exam: A second oral exam in the next quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Elective	Yes
Tags	Mathematics Signals and Systems Stochastics

WI4260TU	**Scientific Programming for Engineers**	**3**
Responsible Instructor	Prof.dr.ir. H.X. Lin	
Practical Coordinator	Ir. C.W.J. Lemmens	
Contact Hours / Week x/x/x/x	0/0/2/0 + lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: experience with a programming language (e.g., Python or C), and linear algebra Strongly advised: experience with linux is a plus (but not a prerequisite).	
Course Contents	The course tries to bring students to a level where they are able to change algorithms from e.g., numerical analysis into efficient and robust programs. It comprises 1. Introduction to programming in general; 2. Floating point number rounding-off errors and numerical stability; 3. (Numerical) Software design; 4. Data Structures; 5. Testing, debugging, and profiling; 6. Efficiency issues in computing time and memory usage; 7. Optimization and dynamic memory allocation; 8. Scientific software sources and libraries. P.S. (i) This is not a general introductory course to learn how to write a computer program. Furthermore, the course concentrates mainly on sequential programming and only briefly introduces parallel programming (MPI and OpenMP). More advanced topics like threads or parallel (MPI/GPU) programming on supercomputers are not covered by this course (they are covered by other courses, e.g., Introduction to High-Performance Computing). (ii) This is a course aimed at master students, however, if you have followed the course Scientific Programming in the BSc CSE-minor, then you should not select this course as a Master elective course due to the large overlap.	
Study Goals	1. Is able to explain the problems related to the use of floating-point numbers in scientific computing; 2. Is able to implement, compile, and debug programs in a Linux programming environment, e.g., use of a makefile. 3. Is able to explain the impact of data structures on the performance of a program, e.g., data locality, cache and memory access. 4. 3. Is able to explain the impact of the program on the performance of a program, e.g., temporal locality, pipelining, and loop-carried dependency. 5. Is able to analyze the performance of a program, and can apply optimization, debugging, and profiling tools to improve of the program. 6. Is able to make the transition from a scientific model to a structured program;	
Education Method	Weekly there are 2-hour lectures and 3-hour lab sessions. Participation in a minimum of five lab sessions is required, with the optional homework, a bonus of max. 1 point can be obtained.	
Books	Textbook: Writing Scientific Software - A guide to good style, by Suely Oliveira and David Stewart, Cambridge University Press, ISBN-13 978-0-521-67595-6	
Assessment	1. Written exam: 3-hour exam comprising theory and practical (lab) questions. (90%) 2. Lab homework (10%) Final grade calculation: (0.9*Exam + 0.1*Lab homework) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: resit in next quarter - Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see TER Art 29, sub 4).	
Tags	Numeric Methods Programming	
Co-Instructor	Dr. D. Palagin	

WI4620	Semidefinite Optimization	6
Responsible Instructor	Dr. D. de Laat	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Semidefinite programming is a generalization of linear programming in which we optimize a linear function over positive semidefinite matrices subject to linear constraints. Optimizing over positive semidefinite matrices adds a measure of nonlinearity while preserving tractability. Semidefinite programming often provides superior modeling power when applying it to more theoretical problems in mathematics and theoretical computer science, while still being efficiently solvable in theory and practice. It is nowadays the central technique in nonlinear convex optimization, finding applications in various fields including control theory, combinatorial optimization, theoretical computer science, discrete geometry, and quantum information theory. In this course, we study semidefinite programming fundamentals and use semidefinite programs to model various problems in mathematics and theoretical computer science. We start by discussing the fundamentals of the positive semidefinite matric cone and semidefinite programs. Then we discuss conic programming duality, for which we will find many applications later in the course. In the remainder of the course, we apply semidefinite programming to various problems, including an application in quantum information (no prior quantum knowledge required) and in machine learning (no prior machine learning knowledge assumed).	
Study Goals	* Derive and apply fundamental properties of the positive semidefinite matrix cone and positive semidefinite programs. * Apply the basic tools and theorems of conic programming. * Formulate and work with the dual of a semidefinite program. * Apply semidefinite programming to the max-cut problem and Grothendieck's inequality. * Apply semidefinite programming to nonlocal games in a quantum information context (no prior quantum information knowledge assumed). * Derive bounds for combinatorial problems using semidefinite programming. * Apply symmetry reduction to simplify semidefinite programs. * Prove results about sums-of-squares characterizations. * Model problems involving polynomial inequality constraints using sums-of-squares characterizations. * Apply semidefinite programming to a machine learning problem (no prior machine learning knowledge assumed).	
Education Method	The course consists of one blackboard lecture each week. The notes used in the lecture will be uploaded to Brightspace.	
Assessment	The assessment is a (closed-book) written exam. The same holds for the re-exam.	
Co-Instructor	Dr. F.M. de Oliveira Filho	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

S&C YEAR 2

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

S&C Graduation Project Obligatory (45 ECTS)

SC52010	S&C Literature Research	10
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The literature study is typically the initial phase of an MSc Thesis project (SC52045), with the purpose to get acquainted with the scientific publications within the realm of the MSc-thesis project, and to prepare for the specific topics to be investigated. The student will need to search for recent publications (i.e. articles, theses, books) that are relevant for the particular thesis project. It is important to be very careful in judging the literature, since not everything written even in high-standard journals is useful - or even correct. In other words, one should be very critical and selective of which publications to use, and one should try to fully understand those that are relevant. See also http://www.dcsc.tudelft.nl/Education/--> Graduation Guide for guidelines to perform literature searches. Moreover the student needs to identify the current issues in the specific research area in order to avoid performing research on questions that have already been resolved in the literature. Once some well-motivated choices have been made on what is planned to be to investigated, these are summarized in a report. This will then form the basis for the subsequent MSc-project work.	
Study Goals	The study goals of the literature study are: 1. Find and study efficiently scientific literature related to the research topic 2. Collect, understand, and familiarize with applicable methods 3. Formalize and finalize the problem statement based on the state-of-the-art literature and terminology 4. Write a scientific report in the form of a literature survey that presents a coherent overview of relevant methods and how they are related to the research question	
Education Method	The literature study is a self study, under supervision of your thesis supervisor.	
Assessment	The literature report is assessed by the thesis supervisor. Important aspects that are taken into consideration are the contents, the organization and clarity in writing and also the process in which the study is performed. An evaluation form is available from the DCSC website.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC52135	S&C MSc Thesis	35
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The MSc-thesis work is the final assignment in the MSc-program, during which a student either further develops the theoretical knowledge gained in the literature survey, or applies it in the form of computer simulations or in the form of experiments (depending on the chosen project). The student will set up and carry out a research project in the field of Systems and Control. The subject of research can be provided by the staff or a company, however it is also possible to propose another project. The final project can be carried out within the framework of ongoing research at the university, within a company, within a research institute or at another university. See also https://www.tudelft.nl/en/me/about/departments/delft-center-for-systems-and-control/education/ . This project also encompasses 2 formal colloquium presentations (Mid-term colloquium and Final MSc colloquium). The students have to attend 15 student colloquia (including those where they give their own presentations). Attendance is monitored. Colloquia sessions are organized throughout the year and are published on the DCSC website and communicated to 2nd year students by email. Workshops with fellow students will be organized by individual staff members or by several staff members together. Your thesis supervisor will discuss this with you. Every S&C MSc Thesis project needs to have an MSc advisor involved who approves the project, serves in the final exam committee, and is a staff member of DCSC .The student may receive daily supervision from a DCSC staff member, a staff member from another department/faculty, a company supervisor (if applicable), or from a PhD student. If a student undertakes the graduation project at a company, an additional graduation agreement should be signed. In this respect, the template of the TUD-ME graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/me-student-portal/education/related/student-forms/msc-forms.	
Study Goals	1. Formulate a formal/mathematical description of the problem setup 2. Describe and reflect on alternative formulations, establish fallback options (contingency planning) 3. Construct a research plan to be executed as part of the thesis project, and specify further details of the research objectives 4. Propose at least one approach as a potential solution to solve the problem 5. Mid-term colloquium: Present the detailed problem formulation and a proposed approach in front of an audience 6. Develop new methods, or extend/modify/compare existing methods in a novel way to address the research question 7. Implement, evaluate, and if needed, perform several (re-)design cycles of the developed techniques either in numerical simulation or based on an experimental setup 8. Engage in scientific discussions with peers (attend colloquia) 9. Write a scientific report (thesis) presenting and discussing the motivation, problem setup, approach, and results/conclusions. 10. Present an overview of the developed methods, the obtained results and research findings, together with the corresponding conclusions in front of an audience (final presentation) 11. Defend the performed research and its findings in front of a committee of experts.	
Education Method	Self study with regular supervision from staff members. Participation in colloquia and workshops.	
Assessment	Written report, oral presentations (mid-term and final), oral defense, research process evaluation.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	If an SC Research Assignment (SC52055) was completed before, then the emphasis will be on the last 6 study goals, and no additional mid-term colloquium is needed.	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Systems and Control**

S&C Elective Course Select 1 out of 3: joint interdisciplinary project, or research project + min 5 EC additional electives, or min 15 EC additional electives

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

SC52055	**Research Assignment**	**10**
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All S&C obligatory courses finished.	
Course Contents	The research assignment can be either a theoretical, design, experimental or programming/simulation assignment on a wide range of topics often related to ongoing research. The subject for the assignment will be chosen together with the thesis supervisor, (who will also be the supervisor for the research assignment).	
Study Goals	1. Formulate a formal/mathematical description of the problem setup 2. Describe and reflect on alternative formulations, establish fallback options (contingency planning) 3. Construct a research plan to be executed as part of the thesis project, and specify further details of the research objectives 4. Propose at least one approach as a potential solution to solve the problem 5. Mid-term colloquium: Present the detailed problem formulation and a proposed approach in front of an audience	
Education Method	Self study with regular supervision.	
Assessment	Assignments	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	The mid-term colloquium grade will be taken into account in the final grade, see grading rubric.	
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